TRAINING FOR LONG COURSE WITH A FULL LIFE
COMPLETE Speaker Guide – ALL 79 SLIDES

SLIDE 1 – Title & Welcome
OPEN WITH:
"I'm Angela Naeth. You think you need 20+ hours/week to do long-course. That's wrong."
SAY:
- By end: real hours (8–15h/week), zones, weekly system, 7-day action plan.
- Website: angelanaethcoaching.com

EXPAND:
- I'm a professional triathlete and coach working with age-groupers (not just elites).
- Most athletes heard the 20-hour myth from magazines and Instagram pros.
- But those are elites training for Kona or world championships, paid to train, zero outside stress.
- You're working full-time, raising a family, balancing life.
- Real number: 8–15 hours/week. That fits.
- By end: you'll know exactly what training looks like, how to structure it, how to protect family time, how to start this week.

SLIDE 2 – Roadmap
SAY:
- 6 modules: Reality Check → Science → Weekly System → Life Happens → Consistency → Action Plan.

EXPAND:
- **Module 1 (Reality Check)**: Breaking the myth. Real numbers. Real examples. Family conversation script.
- **Module 2 (Science)**: Testing protocols. Zones. How to calibrate training so you know exactly what "easy" and "hard" mean.
- **Module 3 (Weekly System)**: How to structure your week. Four anchor sessions. Three training levels. Real examples.
- **Module 4 (When Life Happens)**: Because life WILL happen. Deadlines. Family emergencies. Illness. You'll have a Plan B.
- **Module 5 (Consistency & Confidence)**: How to read your body. When to push. When to pull back. How confidence is built through familiarity.
- **Module 6 (Action Plan)**: Concrete next steps. Your 7-day blueprint. How to reach out for support.

SLIDE 3 – Reality of Long Course Training
SAY:
- Most finishers: 70.3 = 8–12 h/wk, Ironman = 10–15 h/wk (not 20+).
- Three keys: consistency, smart execution, flexible plan.
- Timeline: 6 months min, 9–12 months ideal (but you're 8–12 weeks out, so we adapt).

EXPAND:
- "Most finishers train 8–15 hours per week." That's data, not opinion.
- Where does the 20-hour myth come from? Elite training. Elite athletes training for world championships do 20–30 hours/week.
- But they're paid to train. They have recovery specialists, massages, perfect sleep, zero job stress.
- You don't have those luxuries. And you don't need them.
- What you need: consistency (showing up week after week), smart execution (knowing your zones, pacing correctly, fueling properly), and flexibility (Plan A for calm weeks, Plan B for crazy weeks).

- Timeline: if you're 8–12 weeks out from your race, you can do this. Minimum 6 months is ideal for someone starting from scratch, but 8–12 weeks is the reality for most of you here.
- Why? Because you're here NOW. You've committed to a race. We're going to make it work.

SLIDE 4 – Module 01 Overview
SAY:
- Compare 70.3 vs Ironman time commitment.
- Debunk 20-hour myth.
- Present athlete examples.
- Timeline.
- Family talk.
- Core philosophy (repeatable, recoverable, progressive).

EXPAND:
- Module 1 covers four things: distance options, myth-busting, real examples, and family buy-in.
- Why two distances matter: different time commitments, different race experiences, both doable.
- 70.3: faster race day (5–7h), less recovery time, 2–3 races/year possible.
- Ironman: longer race day (10–17h), deeper commitment, typically one race/year.
- The myth persists because elites train 20–30h/week and that's the visible image people see.
- But most finishers? 8–15h/week. That's the real data.
- Sarah, Mike, Elena—these aren't exceptions. They're typical of successful, balanced athletes.
- The timeline: minimum 6 months, ideal 9–12 months. Rushing leads to injury and burnout.
- The family talk: this is critical. Demand for support fails; collaboration works.
- Core philosophy: every week must be repeatable (fits your life), recoverable (you can absorb it), and progressive (moves you forward).

SLIDE 5 – 70.3 vs Ironman
SAY:
- **70.3**: 8–12 h/wk, fits full-time job, 2–3 races/yr, 5–7h race day.
- **Ironman**: 10–15 h/wk, higher priority, 1 race/yr, 10–17h race day.
- Which excites you more?

EXPAND:
70.3 (Half Ironman):
- Weekly time: 8–12 hours/week. Manageable while working and family time.
- Why choose it: shorter commitment window (8–12 weeks prep), faster feedback loop (race soon), less family impact, fun to do 2–3/year.
- Race day: 5–7 hours (typical). Swim 1.2 mi, bike 56 mi, run 13.1 mi.
- Recovery: 7–10 days post-race, then back to normal life.

Ironman (Full 140.6):
- Weekly time: 10–15 hours/week. Larger life commitment, less weekly flexibility.
- Why choose it: deeper challenge, one big goal, ultimate endurance test, possible qualification (Worlds).
- Race day: 10–17 hours (typical). Swim 2.4 mi, bike 112 mi, run 26.2 mi.
- Recovery: 2–3 weeks post-race (physical and mental).

Key insight: Both fit a full life IF you structure training right. Neither requires sacrificing your job or family. It's about design.

SLIDE 6 – The Myth
SAY:
- Myth: 20+ hrs/wk. Reality: 10–15 hrs/wk (most finishers).
- Three pillars: consistency > intensity, smart execution (zones/pacing/fueling), flexible plan.

EXPAND:
The Myth:
- "You need 20+ hours per week to finish an Ironman."

- Where it comes from: magazine articles, Instagram, pro athlete content.
- Why it persists: elites are visible, their training gets shared, it looks impressive.
- Why it's wrong: pros train 20–30h/week because they're paid to train. They have no job. Zero outside stress. Perfect recovery.

The Reality:
- Ironman: 10–15 hours/week (most finishers 12–14h).
- 70.3: 8–12 hours/week (most finishers 10h).
- This is repeatable while working 40h/week and raising a family.
- This is what WORKS for people with full lives.

The Three Pillars:
1. **Consistency over intensity** – A "good enough" week done 12 times beats a "perfect" week done once. 52 consistent weeks beat 4 perfect weeks + 48 off weeks.
2. **Smart execution** – Knowing your zones, pacing correctly (not too hard early), fueling properly (40% of the race right there). This is training GPS.
3. **Flexible plan** – Plan A for calm weeks (full training). Plan B for busy weeks (protect anchors, cut filler). Both move you forward.

Why this matters: You can do this. It's not a fantasy. It fits your actual life.

SLIDE 7 – Real Examples (Sarah, Mike, Elena)

SAY:
- **Sarah (lawyer)**: 12 h/wk, Ironman 12:45, no missed work.
- **Mike (dad of 3)**: 8 h/wk, 70.3 PR by 20m, pre-dawn sessions.
- **Elena (owner)**: 9–12 h/wk variable, Worlds qual.
- All succeeded because they adapted to life.

EXPAND:

Sarah – Corporate Lawyer
- Training: 12 hours/week (consistent, early mornings).
- Race result: Ironman 12h 45m (strong finish).
- Life constraints: full-time law job, high-stress workload, no missed work, kids, partner.
- Strategy: early mornings (5:30 am starts), lunch-break swims, weekends protected for long sessions.
- What worked: family talk first (kids knew Saturday was long-ride day), anchors locked in, Plan B ready when work crises happened.
- Lesson: structure replaces decision-making. Predictable routine reduces friction.
- Quote: "I couldn't do this without my family's support. When I told them the training times and stuck to them, they got on board."

Mike – Dad of Three (Under 6)
- Training: 8 hours/week (minimal, highly optimized, pre-dawn).
- Race result: 70.3 PR by 20 minutes (year-over-year improvement).
- Life constraints: three young kids, wife working, limited family time to protect, chaos at home.
- Strategy: up at 4:30 am, done by 6:30 am before kids wake, bike commute counts, running with stroller counts.
- What worked: quality over quantity, no guilt about Plan B weeks, trained with kids when possible, integration not isolation.
- Lesson: constraints force efficiency. Less time breeds better focus.
- Quote: "I don't have 15 hours/week. I have 8. But those 8 are dialed in. No waste."

Elena – Business Owner
- Training: 9–12 hours/week (highly variable, flexible).
- Race result: Ironman World Championship qualification (multiple times, consistent execution).
- Life constraints: unpredictable schedule, business ebbs and flows, travel monthly, no consistency.
- Strategy: Plan B as default, anchors still protected, willing to adjust blocks based on work cycles.

- What worked: long-term thinking (9-month build, not 16-week panic), trusted the system, adjusted weeks but not principles.
- Lesson: adaptation beats rigidity. The system survives variable conditions if principles are solid.
- Quote: “My schedule changes every week. But I always protect Saturday long bike and Sunday long run. That’s non-negotiable.”

Common Thread: All three protected anchor sessions. All did 8–15 h/week. All adapted to reality instead of forcing perfection. All succeeded.

SLIDE 8 – Your First Long Course (Timeline)
SAY:
- Don’t rush. 6 months minimum, 9–12 months ideal.
- Phase 1 (START): 6–8h/wk, easy, habit-building.
- Phase 2 (BUILD): 9–15h/wk, volume ↑.
- Phase 3 (PEAK): intensity ↑, race sims.
- Phase 4 (RACE WEEK): taper, trust.

EXPAND:
Phase 1 – START (Weeks 1–6)
- Focus: Build routine, establish consistency, develop aerobic base.
- Weekly volume: 6–8 hours/week, mostly easy (Z1, recovery).
- Main goal: habit formation, not fitness yet.
- What you’re doing: 3–4 sessions/week (mostly Z1), one long bike (90–120 min), one long run (45–60 min), easy swims.
- Why this phase: your body is learning to tolerate training stress. Adaptation happens slowly.
- Success marker: if you can repeat this week without falling apart, you’re ready for Phase 2.

Phase 2 – BUILD (Weeks 7–22)
- Focus: increase volume, introduce threshold work, build durability.
- Weekly volume: gradually 9–15 hours/week, 90% easy, 10% hard.
- Main goal: aerobic capacity and sustainable speed.
- What you’re doing: longer rides (2–4h Z1–Z2), longer runs (75–120 min Z1–Z2), one quality session (threshold or VO_2), brick training.
- Why this phase: aerobic adaptations are occurring (mitochondrial growth, capillary expansion).
- Success marker: long sessions feel steadier, recovery is happening, power/pace is increasing slightly.

Phase 3 – PEAK (Weeks 23–28)
- Focus: sharpen, race simulation, practice pacing/fueling.
- Weekly volume: 12–15 hours (slightly down from build, but intensity up).
- Main goal: race readiness, execution confidence.
- What you’re doing: race-pace work, full brick sessions (2h bike + 40m run for 70.3; 3h bike + 60m run for IM), one long session at race intensity, fueling practice locked in.
- Why this phase: you’re proving to yourself that you can execute the race plan.
- Success marker: race-pace efforts feel controlled, nutrition is dialed, confidence is high.

Phase 4 – RACE WEEK (Weeks 29–30)
- Focus: taper, rest, trust the work.
- Weekly volume: 50% of normal (6–8h/week for IM, 4–6h for 70.3).
- Main goal: arrive fresh, rested, ready.
- What you’re doing: easy spins, short quality burst (no new workouts), lots of sleep, no long sessions, movement only.
- Why this phase: allow your body to recover and absorb all the training.
- Success marker: you feel fresh (not stale), you’re sleeping well, you’re excited (not terrified).

Key insight: “Rushing this timeline is where most people fail. Give yourself the runway. Trust the process.”

SLIDE 9 – The Family Talk

SAY:

- Script: "I want to race, not at our expense. Can we plan this together?"
- Set untouchable family time zones (e.g., Sat 6–9am ride, rest is ours).
- Ask: "When could you support a training week?"

EXPAND:

Why The Family Talk Matters:

- Training doesn't happen in a vacuum. It happens in a family, with a partner, with kids, with responsibilities.
- If your family doesn't understand and support this, you'll face resentment, guilt, and ultimately failure.
- Resentment is the #1 reason people quit training. Guilt kills consistency.
- This conversation prevents the slow burn of passive-aggressive comments, missed family events, and growing distance.

Before You Start Training (The Opener):

- Choose a calm moment (not stressed, not rushed).
- Sit down (not while cooking, not while checking email).
- Script: "I want to train for [race name] on [date]. It's important to me. But not at our expense. I want us to figure this out together. What questions do you have? What concerns you?"
- Listen. Actually listen. Don't defend.
- Example concerns: "Will you miss family time?" "Are we putting our weekend on hold?" "Is this safe?" "Why do you want to do this?"
- Address each concern honestly.

During Training (The Ongoing Conversation):

- Be specific about your schedule.
- Script: "My long ride ends by 11 am on Saturday. The rest of the weekend is ours."
- Set "untouchable" family time zones (e.g., dinner together 5 nights/week, Saturday afternoon as family, no training on Sunday evening).
- Involve them: bring kids on easy rides, invite them to training (not just race day).
- Show appreciation: "I couldn't do this without your support."

Establishing "Untouchable" Zones:

- Example 1: Saturday 6–9 am (long ride), rest of weekend family-focused.
- Example 2: Weekday 5:30–6:30 am (swim/run), home by 6:45 for breakfast with family.
- Example 3: No training Friday night (date night, family connection).
- The key: they know exactly when you're training and when you're theirs. No surprises.

Make it Collaborative:

- Ask: "When could you support a training week? What days work? What days don't?"
- Don't dictate; negotiate.
- Show how training integrates: "Can the kids bike along the easy run?" "Can Saturday long ride end at coffee shop for brunch?"
- Win-win: training fits, family feels included.

Red flags (if these appear, revisit the conversation):

- Your partner makes comments about "your races again."
- Kids avoid asking about training (shame).
- Family events get cancelled or postponed because of training.
- You feel guilty every session.
- Your partner says, "I supported you in the beginning, but now…"

If that's happening: Step back, adjust training level, rekindle the family conversation. Consistency > perfection; family > racing.

SLIDE 10 – Core Philosophy (Repeatable, Recoverable, Progressive)

SAY:

- Three things make training stick: **repeatable** (fits your life), **recoverable** (you can absorb it), **progressive** (moves you forward).
- Consistency > perfection. One good week done 12 times beats one perfect week.

EXPAND:

1. Repeatable: Fits Your Lifestyle

- Can you do this week again next week? And the week after?
- If the answer is "only if everything goes perfectly," it's not repeatable.
- If the answer is "yes, even if my work spikes or kids get sick," it is.
- Repeatable = realistic. Not fantasy training; actual training.
- Example repeatable week: 12 hours, early mornings, lunch-break swims, weekend anchors, Plan B ready.
- Example non-repeatable week: 18 hours, heroic efforts, perfect sleep, zero life stress, no flexibility.

2. Recoverable: You Can Absorb It

- Your body doesn't adapt during training; it adapts during recovery.
- If a week leaves you cooked (TSB below -20, deep fatigue, can't sleep), it's not recoverable.
- If a week leaves you tired but still functional, it is recoverable.
- Recoverable depends on: sleep quality, nutrition quality, life stress, stress management.
- If life stress is high (deadline at work, family crisis), your body can't recover from even a normal training week—so training volume must drop.
- The stress budget is real: training + work + family + sleep loss = total stress. Keep it within your recovery window.

3. Progressive: Moves You Forward

- Each week (or each month) should be slightly harder or longer than the last.
- Progression = increased volume, increased intensity, increased durability, improved power/pace/swim speed.
- But progression is slow. 5–10% per week, not 50%.
- Progressive doesn't mean "always increasing." It means "over 4–6 weeks, volume/intensity is trending up."
- If you're doing the same 12 hours/week for 12 weeks, you're not progressing (stagnation).
- If you're spiking from 10h to 16h in one week, you're not progressing (you're risking injury).
- Progressive = steady forward movement, managed growth.

Why Consistency > Perfection:

- A "good" week (10–12 hours, solid execution) done 12 times = huge fitness.
- A "perfect" week (20 hours, heroic efforts, flawless execution) done 3 times + missing weeks due to burnout = no fitness.
- Consistency compounds. 52 good weeks (one per year) beats 4 perfect weeks + 48 off weeks.
- Athletes who finish strong: consistent. Athletes who get injured or plateau: perfect-then-crash cycles.

The Athlete Mindset:

- "Build a body that handles consistent stress, not just one crushing workout."
- This means: your goal isn't one perfect week; it's 24 solid weeks.
- It means: boring is good. Predictable is good. Steady is good.
- It means: if you miss a week, you don't panic and do a hero week next week; you resume gently.

The Proof:

- Sarah (12h/week, lawyer): finished Ironman strong because she repeated 12h/week 24 times, not because of one 20h week.
- Mike (8h/week, dad): PR'd 70.3 because he executed 8h/week consistently, not because of a single monster week.
- Elena (9–12h/week, business owner): qualified for Worlds because she was repeatable over 36 weeks, adapting but not breaking.

SLIDE 11 – Module 02: The Science That Matters

SAY:

- Stop guessing. Start calibrating.
- Testing gives you zones. Zones drive training. Zones = progress.

EXPAND:

- Most people train blind—they guess at effort, they follow a generic plan, they hope it works.
- We're not doing that. We're going to measure your fitness, identify your zones, and train with data.
- This takes 2–3 hours to test, but the payoff is 12+ weeks of precise training.
- Here's why: without zones, you probably train in the "gray zone" (moderately hard, no recovery, no real intensity). That's the worst place to train.
- Smart athletes train 90% easy (foundation building) and 10% hard (walls and roof).
- Testing creates zones: you know exactly what "easy" and "hard" mean in your body, in watts, in pace, in meters/minute.
- Without zones: you guess at effort and probably go too hard on recovery days and not hard enough on hard days.
- With zones: every workout has a purpose, every session targets a specific stimulus, every week builds methodically.

SLIDE 12 – Why Testing Matters (Knowing Your Zones)
SAY:

- Testing creates clarity.
- Three reasons: target exact zones, track progress, confirm feeling vs reality.
- Testing = calibration.

EXPAND:
Reason 1 – Target Exact Zones:

- Without testing: you have no zones. You run "medium hard" or "easy-ish" (vague, useless).
- With testing: you know exactly what 70% of your power looks like. You know your threshold pace. You know your swim speed for different efforts.
- Exact zones ensure correct physiological stimulus every session.
- Example: If your threshold is 280W and you do a Z2 workout, you know to ride 210–252W (75–90% threshold). Not 280W (that's threshold, not Z2). Not 150W (that's Z1, not Z2).
- Precision = adaptation. If you miss the zone by 10%, you miss the stimulus and waste the session.

Reason 2 – Track Progress:

- Without testing: you don't know if you're improving. You guess: "I feel stronger?"
- With testing: you retest every 8–12 weeks. Your power goes from 280W to 300W. Your 5k pace goes from 7:20 to 7:05. Your swim 400m time goes from 5:40 to 5:25. That's objective progress.
- Seeing progress is motivating. Knowing you're improving drives consistency.
- Objective data beats subjective feeling. On a bad day, you feel slow; the data might show you're still improving. That's encouraging.

Reason 3 – Confirm Feeling vs Reality:

- Without testing: your perception might be wrong. You feel easy, but maybe you're actually coasting. You feel hard, but maybe that's not max effort.
- With testing: you validate RPE (rate of perceived exertion) with data. Your zone 1 should feel conversational; if the data says zone 1 but you're gasping, your perception is off. Or the data is off.
- Over time, testing teaches your body to read effort correctly. You build the mind-body connection that separates smart athletes from hard-workers.

Testing = Calibration:

- You're not guessing anymore. You're calibrating your system.
- Every session has a purpose. Every zone is targeted. Every week builds on the last.
- This is why testing athletes progress faster than non-testing athletes.

SLIDE 13 – Start Testing Now (3 Reasons Recap Visual)
SAY:

- Three checkmarks: target exact zones, track progress, confirm feeling.
- Testing stops guesswork. Creates clarity. Drives progress.

- If you take one thing from today: test your zones before you start training. Don't guess.

EXPAND:

- The three checkmarks represent three reasons: target exact zones, track progress, confirm feeling.
- These aren't nice-to-haves; they're essential for efficient training.
- Most athletes skip testing because "it takes time" or "I know my effort." Big mistake.
- 2–3 hours of testing saves 50+ hours of wasted training.
- Testing happens early in your build (week 1–2 ideally, before you lock in zones for 12 weeks).
- You'll retest at week 12 or 16 (depending on block) to see progress and update zones.
- The mantra: "Stop guessing and start using data to drive your training decisions."

SLIDE 14 – Zones Build the House

SAY:

- Foundation (Z1): 60–70% training, durability.
- Walls (Z2–Z3): 20–25% training, threshold.
- Roof (VO_2): 5–10% training, peak power.
- Build bottom up. Skip steps = collapse.

EXPAND:

Layer 1 – Foundation (Zone 1 + Recovery)

- **What it is**: easy effort, you can hold a conversation, aerobic base building.
- **% of training**: 60–70% should be here.
- **Why it matters**: all fitness is built on this. Without a strong foundation, harder training creates injury, not adaptation.
- **Session examples**: easy spins on the bike, easy jogs, easy swims, recovery rides/runs.
- **Duration**: 30–90 min, multiple times per week, can be done daily without fatigue.
- **Feel**: conversational, sustainable, you could do this all day.

Layer 2 – Walls (Zone 2 and Zone 3)

- **What it is**: harder sustained effort, threshold pace, you can talk in short sentences but not paragraphs.
- **% of training**: 20–25% should be here (split between Z2 and Z3).
- **Why it matters**: this is where "speed" lives. This is where you build the ability to sustain hard effort.
- **Session examples**: threshold rides, tempo runs, brick sessions, longer Z2 holds.
- **Duration**: 45–90 min for Z2, 20–40 min for Z3, once or twice per week max.
- **Feel**: uncomfortable but controlled, you can hold this for the duration, afterward you need recovery.

Layer 3 – Roof (VO_2 Max)

- **What it is**: high-intensity effort, near-max power/pace, you can only hold for short intervals.
- **% of training**: 5–10% should be here.
- **Why it matters**: this is where peak power and speed are trained. This is the "ceiling" of your fitness.
- **Session examples**: 5-min hard efforts, 1-min intervals at VO_2, max-effort swims.
- **Duration**: 3–8 min intervals (repeats), 10–20 min total session time, once per week max.
- **Feel**: very hard, breathing heavy, can only hold for the prescribed time, recovery is necessary.

The Critical Insight:

- Most athletes train too much on the roof (doing hard workouts every session) and neglect the foundation.
- This creates: quick burnout, injury, inability to hold hard effort for long (critical on long-course racing).
- Elite long-course athletes train 90% easy, 10% hard. That ratio works.
- This is counterintuitive. You'd think more hard = more fitness. But for endurance, more easy + strategic hard = more fitness.

Why This Matters for Long-Course Racing:

- 70.3 and Ironman are races of durability, not peak power. You need a massive aerobic base (foundation) to survive 5–17 hours.
- You need threshold capacity (walls) to hold race pace for that duration.
- You don't need peak VO_2 (roof) for long-course racing—you need it for short, intense races.
- This means: your training blocks should emphasize foundation and walls, with periodic roof work, not the other way around.

SLIDE 15–16 – Bike & Run Training Zones (Visuals)
SAY:
- Visual of zones from recovery to VO_2.
- Recovery < Z1 < Z2 < Z3 < VO_2.
- Same concept for both.

EXPAND:
- Run zones are pace-based, not power-based (unless you have a power meter).
- Recovery zone run pace: conversational, building aerobic base without fatigue.
- Zone 1: easy run pace, you can talk in full sentences, most of your weekly running is here.
- Zone 2: tempo run pace, harder but sustainable, 20–40 min at a time.
- Zone 3: threshold pace, very hard, 10–20 min at a time, feels like you're working hard.
- VO_2 max: fast intervals, 3–8 min repeats, feels hard.
- The visuals show the progression: more time at bottom (easy), progressively less time at top (hard).
- This reflects real training: 60–70% easy, 20–25% threshold, 5–10% VO_2.

SLIDE 17 – Critical Power and Speed (The Anchor)
SAY:
- CP = power you can hold ~30–60 min (threshold).
- CS = pace you can hold ~30–60 min (run).
- More accurate than FTP or heart rate.
- Your zones are % of CP/CS.

EXPAND:
What is Critical Power (CP)?
- The power output you can theoretically sustain for about 30–60 minutes at max effort.
- More accurate than FTP because it's calculated from two all-out efforts (3-min and 12-min), not guessed.
- Your zones are percentages of CP: Z1 = 60–75% CP, Z2 = 75–90% CP, Z3 = 90–105% CP, VO_2 = >105% CP.
- Example: if your CP is 250W, then Z2 is 188–225W. Every workout in Z2 targets that range.

What is Critical Speed (CS)?
- The pace you can theoretically sustain for about 30–60 minutes at max effort (running version of CP).
- Calculated from all-out 400m, 800m, and 3200m efforts.
- Your zones are pace bands around CS: easy pace, Z1 pace, Z2 pace, etc.
- Example: if your CS is 7:00/mile, then Z1 is roughly 7:30–8:30/mile, Z2 is roughly 7:00–7:30/mile.

Why CP and CS Are Essential:
- They eliminate guesswork. You know exactly what "hard" and "easy" mean.
- They allow you to train with purpose. Every session targets a specific zone, a specific adaptation.
- They're your training GPS. Without them, you're driving blind.
- They're more accurate than heart rate (HR varies with fatigue, caffeine, sleep, stress) and more reliable than RPE (perceived effort varies with mindset).

Why This Matters:
- Elite coaches use CP and CS because it works.
- Age-group athletes who test CP and CS progress faster than those who guess.

- Testing takes 2–3 hours; training with zones saves 50+ hours of wasted effort.
- The mantra: “Find your CP and CS. Train your zones. Progress will follow.”

SLIDE 18 – Bike & Run Testing Protocols
SAY - BIKE:
- Warm 15 min → 3-min all-out → recovery → 12-min all-out → cool down.
- Record average power both efforts.

SAY - RUN:
- Warm 10 min → 400m all-out, 800m all-out → recovery → 3200m all-out.
- Record times.

SAY - SWIM:
- 400m all-out (T400) → 48h later → 1500m all-out (T1500).

SAY:
- Must be all-out, fully recovered, 48–72h between tests, retest every 8–12 weeks.

EXPAND:
BIKE: Critical Power Testing
Test 1 – 3-Minute and 6-Minute All-Out:
- **Warm-up**: 15–20 minutes of progressive effort (easy → moderate → hard), get your heart rate elevated.
- **3-Minute all-out**: From second 1, you’re going as hard as you can for a full 3 minutes. This is brutal. Record average power.
- **Recovery**: 10 minutes easy pedaling, let heart rate come down.
- **6-Minute all-out**: Same deal—go as hard as you can for 6 minutes. Record average power.
- **Cool down**: 5–10 minutes easy.
- **Equipment needed**: power meter (bike computer or trainer app), flat road or trainer (wind affects power data).
- **Timing**: morning, well-rested, fed, no hard workout the day before.

Test 2 – 12-Minute All-Out:
- **Timing**: 48–72 hours after the 3/6-minute test (fully recovered).
- **Warm-up**: same as above, 15–20 minutes progressive.
- **12-Minute all-out**: go as hard as you can sustain for 12 minutes. Yes, this is hard. Very hard. Record average power.
- **Cool down**: easy pedaling afterward, let body recover.
- **Result**: your average power from this 12-minute effort is close to your critical power.

RUN: Critical Speed Testing
Test 1 – 400m and 800m Time Trials:
- **Warm-up**: 10–15 minutes easy jog, get heart rate elevated, do a few strides.
- **400m all-out**: run a track (or GPS flat route) 400m as fast as you can. Record time.
- **Recovery**: 3–5 minutes easy jog/walk.
- **800m all-out**: same deal, go as hard as you can for 800m. Record time.
- **Cool down**: easy jog home.
- **Equipment needed**: GPS watch or track, flat surface, traction (not slippery).
- **Timing**: morning, well-rested, fed, no hard run the day before.

Test 2 – 3200m Time Trial:
- **Timing**: 48–72 hours after the 400m/800m test (fully recovered).
- **Warm-up**: 10–15 minutes progressive easy jog, get heart rate up.
- **3200m all-out**: run 3200m (2 miles) as hard as you can sustain. This is a tempo effort. Record time.
- **Cool down**: easy jog afterward.
- **Result**: your pace from this 3200m effort, plus your 400m and 800m times, determine your critical speed via a formula (we calculate this in coaching).

Critical Success Factors (For Both Bike and Run):

- **All-out means all-out**. If you hold back, your zones will be wrong and training will be off all season.
- **48–72 hours between tests**. You need full recovery; otherwise, fatigue masks your true fitness.
- **Fully recovered before testing**. No hard workouts in the 3 days before. Sleep well. Eat well. Hydrate.
- **Test every 8–12 weeks**. Your fitness improves; your zones change. Update zones regularly.
- **Input results immediately**. After testing, calculate your zones (use a calculator or coaching app) and lock them in before the next workout.

Testing Logistics:

- Pick a day you can dedicate to testing (morning, fresh, no other demands).
- Do bike test one day, run test 3 days later (or vice versa). Don't test both on the same day.
- Tell your family you're testing; they'll understand. It's 2–3 hours total, spread over a week.
- This is an investment that pays off for 12+ weeks.

SLIDE 19 – Swim Training Zones & Testing (CSS)

SAY:

- Swimming is different (no power meter).
- Use Critical Swim Speed (CSS): pace you can hold ~30–60 min.
- Build zones around CSS.
- Principle: frequency beats single long sessions (four 2.5k swims > one 10k swim).
- Weekly goal: 70.3 = 6–12k/wk, Ironman = 8–15k/wk.

EXPAND:

How to Test CSS (Critical Swim Speed)

Day 1 – 400m All-Out:

- **Warm-up**: 400m easy swim, get body moving, loosen up.
- **400m all-out**: swim 400m (8 lengths of a 50m pool, 16 lengths of a 25m pool) as fast as you can. Record time.
- **Rest**: 3–5 minutes walk around, recover.

Day 2 (48–72 Hours Later) – 1500m All-Out:

- **Warm-up**: 400m easy swim.
- **1500m all-out**: swim 1500m as hard as you can sustain. This is tough. Record time. (This is your 3/4-mile swim pace.)
- **Cool down**: easy swim afterward.

Calculate CSS:

- Your times from 400m and 1500m determine CSS via a formula: CSS = (1500m time – 400m time) / (1500m distance – 400m distance).
- Example: if 400m = 5:40 (5 min 40 sec = 340 sec) and 1500m = 22:00 (22 min = 1320 sec), then CSS = (1320 – 340) / (1500 – 400) = 980 / 1100 = 0:53 per 100m.
- Zones are built around CSS: recovery <85% CSS, Z1 85–92%, Z2 93–97%, VO_2 98–105%+.

Swim Training Zones (After CSS is Calculated):

- **Recovery zone**: <85% CSS, feels easy, technique focus, active recovery.
- **Zone 1**: 85–92% CSS, aerobic pace, conversational, buildable up to 60+ min.
- **Zone 2**: 93–97% CSS, threshold pace, controlled discomfort, 20–40 min repeats.
- **VO_2 max**: 98–105%+ CSS, fast intervals, 50–200m repeats, hard effort.

Key Principle: "Frequency Beats Single Long Sessions"

- Wrong approach: one 4000m swim per week, trying to build aerobic capacity.
- Right approach: four 2500m swims per week, varied paces, building total volume with less fatigue.
- Why? Frequent, shorter sessions allow better quality in each session. One monster swim leaves you destroyed for days.
- Practical example:
- Bad: Monday 4000m = fatigued rest of week, can't do quality bike/run.

- Good: Monday 2500m + Wednesday 2500m + Friday 2500m + Sunday 2500m = 10,000m/week, but each session is fresh, quality is high.

Weekly Swim Volume Targets:

- **70.3**: 6,000–12,000m/week, 3–4 sessions.
- **Ironman**: 8,000–15,000m/week, 4–5 sessions.
- **Distribution**: 60–70% easy (Z1), 20–25% threshold (Z2), 5–10% VO_2 max.

Key Insight:

- Swim CSS zones are treated exactly like bike CP zones and run CS zones.
- The difference: swim is most accessible (easy to control effort in a pool), so most swimmers can practice CSS zones effectively.
- Testing CSS once per 8–12 weeks shows progress.

SLIDE 20 – Your Training House (Visual)

SAY:

- Foundation (easy, 60–70%) = aerobic engine.
- Walls (threshold, 20–25%) = sustainable speed.
- Roof (VO_2, 5–10%) = peak power.
- Build bottom up. Skip steps = collapse.
- Long-course racing lives on foundation + walls, not roof.

EXPAND:

Foundation (Zone 1 + Recovery) – Layer 1:

- Represents 60–70% of your training.
- Built over weeks 1–6 of the base phase.
- Purpose: aerobic capacity, mitochondrial density, capillary growth, fat metabolism, durability.
- Sessions: easy spins, easy runs, easy swims, recovery days.
- Duration: 30–90 minutes, can be done daily.
- Feel: conversational, sustainable, low stress.
- Why it matters: everything else depends on this. A weak foundation means weak walls and roof.

Walls (Zone 2 + Zone 3) – Layer 2:

- Represents 20–25% of your training.
- Built during weeks 7–22 of the build phase.
- Purpose: threshold power/pace, sustainable speed, ability to hold hard effort.
- Sessions: tempo rides/runs, threshold work, brick sessions, longer zone 2 efforts.
- Duration: 45–90 minutes for Z2, 20–40 minutes for Z3, once or twice per week.
- Feel: uncomfortable but controllable, you can finish the session but need recovery.
- Why it matters: this is where race pace lives. Without strong walls, you'll blow up on race day.

Roof (VO_2 Max) – Layer 3:

- Represents 5–10% of your training.
- Built during weeks 23–28 of the specific/peak phase.
- Purpose: peak power, speed, ability to accelerate and respond to surges.
- Sessions: VO_2 intervals (5–8 min repeats), max-effort sprints, fast swim intervals.
- Duration: 3–8 min intervals, 10–20 min total session time, once per week max.
- Feel: very hard, breathing heavy, you can only hold for the prescribed time.
- Why it matters: for long-course racing, the roof is less critical than foundation and walls (you're not doing max-effort sprints in a 70.3), but having a roof means you can accelerate if needed, pass someone, and finish strong.

The Build Sequence:

1. **Start**: build the foundation (6 weeks of easy training).
2. **Build**: add the walls (4–6 weeks of threshold work).
3. **Specific**: raise the roof (2–3 weeks of VO_2 and race-sim work).
4. **Race**: execute on the foundation, hold the walls, deploy the roof as needed.

Why This Structure Matters for Long-Course:

- Ironman is a test of durability (foundation), not speed (roof). If your foundation is weak, you'll bonk at mile 15 of the run.
- 70.3 is slightly more speed-intensive than Ironman, but still durability-first. Foundation, then walls, then roof.
- Racing on a weak foundation = suffering, bonking, walking.
- Racing on a strong foundation + solid walls = controlled effort, steady pace, strong finish.

SLIDE 21 – Full Life House (Foundation + Walls + Roof Integration)
SAY:
- Foundation = habits (sleep, nutrition, stress).
- Walls = time (training fits your actual hours).
- Roof = goals (peak depends on base).
- Build with your life, not against it.

EXPAND:
Foundation = Habits (Sleep, Nutrition, Stress)
- Fitness is built during sleep, not during workouts.
- Workouts are the stimulus; sleep is the adaptation.
- If you're training hard but sleeping 5 hours/night, you're wasting your training.
- Good habits: 7–9 hours sleep, consistent nutrition, stress management.
- Bad habits: irregular sleep, skipped meals, chronic stress unmanaged.
- Key insight: if your foundation habits are weak, even perfect training won't create fitness.

Walls = Time (Training Fits Your Actual Hours)
- You can't train 15 hours/week if you work 50 hours and have three kids.
- The walls are built from available time: 8–12 hours/week if you're busy, 12–15 hours/week if life is calm.
- Smart athletes design training to fit their actual life, not some idealized version.
- Sarah (lawyer) chose 12 h/week because she has limited daylight in winter and limited weekday time.
- Mike (dad) chose 8 h/week because mornings are his only training window.
- Elena (business owner) stays flexible at 9–12 h/week because her schedule varies month to month.
- Key insight: training that doesn't fit your life doesn't stick.

Roof = Goals (Peak Depends on Your Base)
- If your goal is to finish a 70.3 in 5 hours, you need a certain foundation and wall strength.
- If your goal is to PR the run at a 70.3, you need VO_2 work (roof).
- If your goal is to qualify for Worlds, you need a massive aerobic base plus strong walls plus strategic roof work.
- But here's the key: your goal should be realistic for the time/life you have.
- If you have 8 h/week for 12 weeks, your goal is "finish strong," not "PR." Different goal, different training structure.
- Key insight: peak depends on the base. You can't raise the roof if the foundation is weak.

Integration:
- The best athletes build fitness within their life constraints, not in spite of them.
- This means: sleeping well (habit), fitting training into available time (walls), and then setting goals that match (roof).
- Sarah: strong sleep habits + 12 h/week (fits her schedule) + goal to finish strong at Ironman = she crushed it.
- Mike: good sleep despite three kids + 8 h/week (all he has) + goal to PR 70.3 = he nailed it.
- Elena: variable sleep/life + 9–12 h/week (flexible) + goal to qualify for Worlds = she made it.
- Key mantra: "Build with your life, not against it."

SLIDE 22 – CP:VO_2 Ratio (Personalized Block Sequencing)
SAY:

- Ratio < 0.83 = weak threshold → focus Build phase first.
- Ratio 0.83–0.87 = balanced → standard order works.
- Ratio > 0.87 = weak peak power → focus VO_2 early.
- Test reveals your optimal block sequence.

EXPAND:

What is the CP:VO_2 Ratio?

- CP = critical power (threshold).
- VO_2 max power = your max power output.
- Ratio = CP ÷ VO_2 max power.
- Example: if CP = 280W and VO_2 max = 350W, then ratio = 280/350 = 0.8.

Three Ratio Zones:

Anaerobic Profile (Ratio < 0.83):

- You have high peak power but weak threshold.
- Meaning: you're good at short, hard efforts but can't hold hard effort for 30–60 min.
- Training priority: Build threshold first. Do 4–6 weeks of threshold work (Z2, Z3) before VO_2 work.
- Block sequence: Base → Build (threshold focus) → VO_2 Max → Race Prep.
- Why: strengthening threshold will raise your CP, making VO_2 work more effective.

Balanced Profile (Ratio 0.83–0.87):

- You're balanced. Your threshold is proportional to your peak power.
- Meaning: you're a well-rounded athlete; follow the standard training progression.
- Training priority: Standard block sequencing works well for you.
- Block sequence: Base → Build → VO_2 Max → Race Prep (traditional order).
- Why: your physiology supports the textbook progression.

Aerobic Profile (Ratio > 0.87):

- You have a strong engine (high threshold) but limited peak power.
- Meaning: you're good at steady, sustainable effort but not explosive.
- Training priority: Raise ceiling (VO_2) first. Do 2–3 weeks of VO_2 work early, then Build (threshold maintenance), then Race Prep.
- Block sequence: Base → VO_2 Max (early) → Build (maintain threshold) → Race Prep.
- Why: VO_2 work will raise your peak power; then threshold work maintains what you've built.

Why This Matters:

- Generic training plans ignore this. They assume everyone follows Base → Build → VO_2 → Race.
- But personalized training adapts to your physiology.
- If you're anaerobic and do VO_2 work too early, you'll plateau because your threshold is still weak.
- If you're aerobic and skip early VO_2 work, you'll never build peak power for race-day surges.
- The CP:VO_2 ratio solves this by telling you exactly which block to prioritize first.

For Your Race:

- 70.3: usually balanced or aerobic profile benefits most (most 70.3 athletes are endurance-focused, not sprint-focused).
- Ironman: usually aerobic profile (pure endurance, less need for peak power).
- Calculate your ratio after testing; we'll tell you which block sequence is optimal for you.

SLIDE 23 – Module 03: The Weekly System

SAY:

- How to structure 8–15 hours/week so they work.
- Four anchor sessions. Three training levels. Real examples. Brick training.

EXPAND:

- **Module 3 covers**: consistency patterns (4 ways to be consistent), durability principles (frequency > heroics, aerobic volume, smart spacing), block sequencing (traditional and personalized), minimum effective training (8–15 h/week, 90% easy, 10% hard), anchor sessions (4 key workouts each week), real-week examples, brick training protocol, time management, and removing friction.

- Most athletes overthink the system. The system is simple: protect anchors, do easy work, add hard work strategically, repeat.
- This module is your practical blueprint.

SLIDE 24 – Consistency Looks Different (4 Paths, 1 Goal)
SAY:
- Typical Plan (rare): perfect linear progress.
- Life Happens (common): ups, downs, misses (realistic, still works).
- Weekend Warrior (risky): massive spikes, injury prone.
- Steady (ideal): repeatable load week-to-week.
- All four athletes can finish. Adapt to your reality.

EXPAND:
Pattern 1 – Typical Plan (RARE: Perfect Linear Progress)
- **What it looks like**: every week, volume increases smoothly. Week 1 = 8h, Week 2 = 9h, Week 3 = 10h, etc. No dips, no life interruptions, everything goes to plan.
- **Who it is**: not many people. This requires zero life stress, perfect recovery, zero missed workouts.
- **Pros**: clean progression, easy to follow, predictable.
- **Cons**: unrealistic for most athletes. Life happens. Work spikes, kids get sick, sleep dips.
- **Outcome**: if you can maintain this, you'll be very fit. But most people can't, so they get frustrated and quit.
- **Lesson**: don't expect this. Plan for the messy version.

Pattern 2 – "Life Happens" (COMMON: Ups, Downs, Misses)
- **What it looks like**: Weeks 1–3 smooth sailing, Week 4 work deadline, volume drops to Plan B, Week 5 back to normal, Week 6 normal, Week 7 family crisis, missed week, Week 8 return gentle, then continue forward.
- **Who it is**: most age-groupers. Life is unpredictable.
- **Pros**: realistic, adaptable, still moves forward despite interruptions.
- **Cons**: requires flexibility and trust in the system. Hard to stay focused when plans keep changing.
- **Outcome**: progress is slower than "typical plan," but it's sustainable. You finish strong because you adapted.
- **Lesson**: this is your baseline. Expect life interruptions. Have Plan B ready.
- **Real example**: Sarah (lawyer) → March: trial prep, drops to Plan B (keeps anchors, cuts filler); April: back to full schedule, scales volume back up slowly; June: family vacation, another Plan B week; continues building → finishes Ironman strong.

Pattern 3 – "Weekend Warrior" (RISKY: Boom/Bust Spikes)
- **What it looks like**: Mon–Thu almost nothing (life is crazy, no time), Friday = explosion, Saturday = big long session, Sunday = long run, then collapse into nothing again.
- **Who it is**: athletes with unpredictable weekdays, trying to cram training on weekends.
- **Pros**: concentrated training time, weekend focus.
- **Cons**: huge spikes followed by crashes, no consistency, high injury risk, body doesn't adapt (it breaks).
- **Outcome**: fast burnout, injury before race day, or a race where you're not ready because adaptation never happened.
- **Lesson**: avoid this pattern. Consistency is frequency, not volume spikes.

Pattern 4 – "Steady" (IDEAL: Moderate, Frequent, Adaptable)
- **What it looks like**: similar load every week (±10% variance), consistent sessions (5–6 per week), small adjustments when life changes, maintains anchors, cuts filler when needed.
- **Who it is**: athletes who understand sustainability. Sarah chose this. Mike chose this. Elena chose this.
- **Pros**: predictable, sustainable, body adapts steadily, injury risk low, life/training balance is good.

- **Cons**: requires discipline to not spike on good weeks, requires trust in slow progress.
- **Outcome**: steady fitness growth, consistent progress, strong finish.
- **Lesson**: this is the goal pattern. Build toward this.

The Key Insight:

- "All four athletes finished. Adapt the plan to your reality."
- Typical Plan is rare and fragile (one life event breaks it).
- Life Happens is common and workable (expect interruptions, have Plan B).
- Weekend Warrior is risky (spikes cause injury).
- Steady is ideal (consistency + adaptation).
- Most people move between Life Happens and Steady, depending on the week. That's normal.

How to Build Toward "Steady":

1. Start with realistic volume (Level 1 or 2).
2. Protect anchors every week (non-negotiable).
3. Add or cut filler based on life stress.
4. Keep week-to-week variance <10% (if this week is 10h, next week should be 9–11h, not 5h or 15h).
5. Repeat for 12 weeks. You'll be steady.

SLIDE 25 – Durability Explained (Frequency > Heroics)

SAY:

- Three principles:

1. Frequency > heroics (5–6 sessions/wk beats 2 sessions with spikes).
2. Aerobic volume (90% easy builds engine).
3. Smart spacing (hard sessions 48–72h apart).

- Build body that handles consistent stress.

EXPAND:

Principle 1 – Frequency: Small Consistent Doses Beat Random Hero Sessions

- **What it means**: doing 5–6 sessions per week at moderate intensity is better than 2–3 sessions with one massive hero session.
- **Why**: your body adapts to repeated stimulus. If you train 5 days/week, your body learns to recover quickly and handle consistent stress. If you train 2 days/week, your body never learns to handle stress efficiently.
- **Example (Bad)**: Monday rest, Tuesday big 3-hour ride (hero), Wednesday rest, Thursday rest, Friday big long run (hero), Sat/Sun rest. That's spiky, not consistent. Your body is constantly recovering from big efforts.
- **Example (Good)**: Mon easy run, Tues bike quality, Wed easy swim, Thurs easy run, Fri swim, Sat long ride, Sun long run. That's 6 sessions, frequency is high, body learns to absorb it.
- **Key insight**: frequency teaches durability. Your body can handle 5–6 sessions/week at the right intensities better than 2–3 sessions with spikes.

Principle 2 – Aerobic Volume: Build the Engine Without Breaking the Chassis

- **What it means**: most of your training (60–70%) should be easy (Z1, recovery). This builds mitochondrial capacity, capillary growth, fat-burning ability—the engine.
- **Why**: endurance sports are aerobic. A huge aerobic base means you can sustain hard effort longer.
- **Example**: 12 hours/week should be ~8 hours easy + 4 hours hard. Not 6 hours easy + 6 hours hard (which would burn you out).
- **The trap**: most athletes do too much moderate-hard training (gray zone), which is neither good recovery nor true hard work. It just fatigues you.
- **Key insight**: boring easy training builds the biggest engine. The race is won on your foundation, not your roof.

Principle 3 – Smart Spacing: Separate Hard Sessions to Protect Quality

- **What it means**: don't do hard sessions back-to-back. Space them 48–72 hours apart to allow recovery.

- **Why**: a hard session (threshold or VO_2 work) creates neuromuscular fatigue. You need 48–72 hours to recover and do the next hard session with full quality.
- **Example (Bad)**: Monday threshold bike (hard), Tuesday VO_2 run (hard). Both sessions are compromised because you're still fatigued from Monday.
- **Example (Good)**: Monday threshold bike (hard), Tuesday/Wednesday easy, Thursday VO_2 run (hard), Friday/Saturday easy, Sunday long run. Hard sessions are spaced, quality is high.
- **Key insight**: spacing hard sessions means you get better quality (higher power, faster pace) and lower injury risk.

The Goal – Build a Body That Absorbs Training Rather Than Breaking Down:

- Durability is the capacity to repeat the same training stimulus week after week and improve.
- Athletes with high durability: can do 12 hours/week for 24 weeks and get stronger every week.
- Athletes with low durability: do 12 hours/week for 4 weeks and get fatigued, then get injured or overtrained.
- Durability is built through frequency, aerobic base, and smart spacing—not through big sessions.

The Mantra: "Consistency > Intensity"

- A "good" 12h week (frequency, easy base, smart spacing) repeated 12 times beats a "hero" 20h week once.
- Your durability is built in boring weeks, not epic weeks.

SLIDE 26 – Build Together, Don't Skip Steps (Training Blocks)

SAY:

- Base (3–6w, ZR–Z1): foundation.
- Build (4–6w, Z2–Z3): walls.
- VO_2 (2–3w): roof.
- Race (3–5w): execution.
- Reset (1–2w): recovery.
- Don't skip steps.

EXPAND:

Block 1 – Durability (Base Phase, 3–6 weeks)

- **Focus**: aerobic base, consistency, establishing routine.
- **Zones**: ZR, Z1 (easy).
- **Volume**: 6–8 h/week, mostly easy.
- **Sessions**: easy rides, easy runs, easy swims, no hard workouts.
- **Goal**: build the foundation of your training house.
- **Why first**: without a strong base, harder training creates injury, not adaptation.
- **Success marker**: you can repeat 8 hours easy every week without fatigue.

Block 2 – Threshold (Build Phase, 4–6 weeks)

- **Focus**: raise the walls, build sustainable speed.
- **Zones**: Z2, Z3 (threshold and tempo).
- **Volume**: 9–12 h/week, 80% easy + 20% threshold.
- **Sessions**: long rides (Z1–Z2), long runs (Z1–Z2), 1–2 threshold/tempo sessions, brick training.
- **Goal**: teach your body to sustain hard effort.
- **Why second**: you need a base before you can add intensity.
- **Success marker**: threshold efforts feel controlled and repeatable; you can do threshold work and still recover for next session.

Block 3 – VO_2 Max (Specific Phase, 2–3 weeks)

- **Focus**: raise the roof, peak power/speed.
- **Zones**: VO_2 max (>105% CP).
- **Volume**: 10–12 h/week, 85% easy + 15% VO_2 (shorter, higher intensity).
- **Sessions**: VO_2 intervals (5–8 min repeats), shorter sessions, still long runs/rides but some high-intensity surges.
- **Goal**: peak your power and speed before race day.

- **Why third**: VO_2 work should come after threshold is strong (CP is high enough to do VO_2 work).
- **Success marker**: VO_2 intervals feel powerful; you can sustain near-max efforts for prescribed duration.

Block 4 – Execution (Race Phase, 3–5 weeks)

- **Focus**: practice race pace, race simulation, fine-tune fueling.
- **Zones**: race-pace (varies, typically Z2 for Ironman, Z3 for 70.3).
- **Volume**: 8–12 h/week, dropping slightly.
- **Sessions**: long sessions at race pace, full brick sessions (race simulation), fueling practice, taper starting in final week.
- **Goal**: prove to yourself that you can execute the race plan.
- **Why fourth**: you're ready; now prove it.
- **Success marker**: race simulation efforts feel controlled; you finish feeling strong, not destroyed; fueling is dialed.

Block 5 – Recovery (Reset Phase, 1–2 weeks)

- **Focus**: recover post-race, reset.
- **Zones**: ZR, Z1 only.
- **Volume**: 2–4 h/week, very easy.
- **Sessions**: easy spins, easy jogs, walks.
- **Goal**: allow your body and mind to recover.
- **Why last**: after peak training, your body needs reset time.

Why Don't Skip Steps:

- If you skip base and start with threshold: your aerobic base is weak, threshold adaptations won't stick, injury risk is high.
- If you skip threshold and start with VO_2: your threshold is weak, you can't sustain VO_2 work, you get fatigued and overtrained.
- If you skip VO_2 and start with race pace: you peak too early, you're fatigued on race day, you don't finish strong.
- Each step builds on the last. Skip one, and the house collapses.

SLIDE 27 – Typical Block Sequencing (Visual Timeline)
SAY:

- Week 1–6: base (durability, ZR–Z1, 6–8 h/week).
- Week 7–12: build (threshold, Z2–Z3, 9–12 h/week).
- Week 13–15: VO_2 (peak, VO_2 work, 10–12 h/week).
- Week 16–20: race (execution, race pace, 8–12 h/week).
- Week 21: recovery (reset, ZR–Z1, 2–4 h/week).

EXPAND:

- Visual shows progression: volume increases through build, dips slightly in VO_2, dips more in race phase, collapses in recovery.
- The visual also shows intensity (top of bar) increasing from base through VO_2, then maintained through race phase.
- The mantra: foundation first, speed second, peak just before race, then rest and recover.

SLIDE 28 – Traditional vs Individualized Block Sequencing
SAY:

- Traditional: everyone does Base → Build → VO_2 → Race.
- Individualized: based on CP:VO_2 ratio, sequence changes.

EXPAND:
Traditional Approach (One-Size-Fits-All):

- Everyone does the same block order: Base → Build → VO_2 Max → Race → Reset.
- Ignores individual physiology.
- Works for balanced athletes but might not be optimal for extremes.
- Pros: simple, standard, most plans follow this.

- Cons: might not match your strengths or weakness.

Individualized Approach (Based on CP:VO_2 Ratio Testing):

Anaerobic Profile (CP:VO_2 Ratio < 0.83):

- Traditional order: Base → Build → VO_2 Max → Race (might not work).
- Personalized order: Base → Build (longer, threshold-focused) → VO_2 Max → Race.
- Why: your threshold is weak relative to peak power. Spend extra time in Build phase raising CP before doing VO_2 work.
- Example: Instead of 4 weeks in Build, do 6 weeks. Let your threshold catch up.

Balanced Profile (CP:VO_2 Ratio 0.83–0.87):

- Traditional order = Personalized order: Base → Build → VO_2 Max → Race.
- Why: your physiology is balanced; the traditional sequence works well for you.
- Example: 6 weeks base, 5 weeks build, 3 weeks VO_2, 4 weeks race.

Aerobic Profile (CP:VO_2 Ratio > 0.87):

- Traditional order: Base → Build → VO_2 Max → Race (might not work).
- Personalized order: Base → VO_2 Max (early, short) → Build (maintain threshold) → Race.
- Why: your threshold is already strong. Your limitation is peak power. Do VO_2 work earlier to raise ceiling, then maintain threshold in Build phase.
- Example: 6 weeks base, 3 weeks VO_2 (early), 4 weeks build (threshold maintenance), 4 weeks race.

The Key Insight:

- Testing your CP:VO_2 ratio tells us where your weaknesses are.
- We build the training blocks to address those weaknesses.
- This personalization can save you weeks of wasted training and lead to better race-day performance.

Example from Real Athletes:

- Sarah (likely balanced): does traditional Base → Build → VO_2 → Race. Progression is smooth.
- Mike (likely aerobic, endurance-focused 70.3 athlete): might do Base → VO_2 (short) → Build → Race. Gets peak power early, then maintains for race.
- Elena (likely balanced or aerobic): traditional order works, but she personalizes based on her test data.

SLIDE 29 – Personal Block Sequencing (Visualization)

SAY:

- Visual shows three profiles: anaerobic (longer build, shorter VO_2), balanced (standard), aerobic (VO_2 early, longer build).
- Same concept (build from foundation up), but timing is personalized.
- Your CP:VO_2 ratio determines which sequence is optimal for you.

EXPAND:

- **Anaerobic version** (< 0.83 ratio): longer base and build phases (yellow dominates longer), shorter VO_2 phase. Overall timeline might be 20–24 weeks instead of 18 weeks.
- **Balanced version** (0.83–0.87 ratio): standard timeline, roughly equal blocks. 18–20 weeks total.
- **Aerobic version** (> 0.87 ratio): short VO_2 phase early (orange peak early), longer build phase (yellow extends), shorter VO_2 peak. Overall timeline might be 18–20 weeks, but timing is different.
- The visual shows that the total volume shape is similar (base, then build, then taper), but the intensity distribution varies.
- The key: there's not one right answer; there are three, depending on your physiology.

SLIDE 30 – Minimum Effective Training

SAY:

- **70.3**: 8–12 h/wk, 90% easy, 10% hard.
- **Ironman**: 10–15 h/wk, 90% easy, 10% hard.
- Quality > junk. Consistency > heroics.

EXPAND:

70.3 Training (8–12 hours/week):

- **Time commitment**: fits full-time job, family, normal sleep.
- **Fits into**: early mornings before work, lunch breaks, weekends.
- **Distribution**: 90% ZR–Z2 (easy), 10% Z3–VO_2 (hard).
- **Example week**:
- Monday: rest or easy 30-min swim.
- Tuesday: 45-min bike at Z3 (threshold intervals) + easy run 20 min.
- Wednesday: easy 30-min run + easy swim 30 min.
- Thursday: easy bike 45 min (commute) + mobility.
- Friday: easy 20-min run (shakeout).
- Saturday: long bike 2.5–3.5 hours, Z1–Z2.
- Sunday: long run 75–90 min, Z1–Z2.
- **Total**: ~10–11 hours/week.
- **Why 90% easy**: long-course racing is 90% easy pace; training accordingly teaches your body race pace.
- **Why 10% hard**: strategic hard work raises ceiling and prevents stagnation.

Ironman Training (10–15 hours/week):

- **Time commitment**: higher priority, planning required, life needs to support it.
- **Fits into**: early mornings, lunch breaks, weekends, some midweek.
- **Distribution**: 90% ZR–Z2 (easy), 10% Z3–VO_2 (hard).
- **Example week**:
- Monday: rest or very easy 30-min recovery swim.
- Tuesday: 60-min bike at Z2, easy run 30 min post-bike.
- Wednesday: 60-min run at Z1, easy swim 40 min.
- Thursday: 60-min easy bike (commute or trainer), strength.
- Friday: 45-min easy swim, 30-min easy run (double, both short).
- Saturday: long bike 3.5–4.5 hours, Z1–Z2, race-fueling practice.
- Sunday: long run 90–120 min, Z1–Z2.
- **Total**: ~12–14 hours/week.
- **Why 90% easy**: Ironman is a test of durability; easy training builds that.
- **Why 10% hard**: enough to maintain sharpness and prepare for race pace.

Quality > Junk Volume:

- A 12-hour week with 90% easy + 10% hard is better than 12 hours of moderate training.
- Moderate-intensity training (gray zone) is neither recovery nor hard work. It's just fatiguing without adaptation.
- Quality means: easy sessions are genuinely easy (you could ride/run all day at this pace). Hard sessions are genuinely hard (you're near max effort, need recovery).

Consistency > Heroics:

- A 12-hour week repeated 24 times (288 hours total) beats a 20-hour week done 3 times + missing weeks due to burnout (60 hours total).
- Consistency teaches adaptation. Your body learns to handle training stress.
- Heroics feel good in the moment but lead to burnout.

The Minimum Effective Dose for Your Race:

- 70.3: 10 hours/week × 12 weeks = 120 hours of training for race day. That's enough.
- Ironman: 12 hours/week × 20 weeks = 240 hours of training for race day. That's enough.
- More than that is bonus, not necessary. Less than that risks underpreparedness.

SLIDE 31 – Protect the Anchors (4 Non-Negotiable Sessions)

SAY:

- **Long ride**: 2.5–4.5h (Z1–Z2), durability + fueling.
- **Long run**: 75–120m (Z1–Z2), resilience + tough legs.
- **Quality session**: 45–60m (Z2–Z3 or VO_2), sharpness.

- **Easy frequency**: 2–3x (Z1), recovery + volume.
- When life gets crazy: cut everything, keep these four.

EXPAND:

Anchor 1 – Long Ride (Durability & Pacing Practice)

- **Purpose**: build aerobic engine, practice bike pacing, practice bike fueling.
- **Zone**: Z1–Z2 (mostly easy, controlled sustained effort).
- **Duration**: 70.3 = 2.5–3.5 hours; Ironman = 3.5–4.5 hours.
- **Frequency**: once per week, usually Saturday.
- **Why it matters**: most of your race is on the bike (56 km for 70.3, 180 km for Ironman). This session teaches you bike fitness and resilience.
- **Non-negotiable because**: it's the biggest time commitment; it builds the most fitness; you learn how to fuel and pace at that duration.
- **If you had to choose one session**: this is it. Long ride is more important than long run for long-course racing.

Anchor 2 – Long Run (Mental Toughness & Run on Tired Legs)

- **Purpose**: build run fitness, mental toughness, practice running off tired legs (brick effect).
- **Zone**: Z1–Z2 (mostly easy, with controlled effort).
- **Duration**: 70.3 = 75–90 minutes; Ironman = 90–120 minutes.
- **Frequency**: once per week, usually Sunday (24 hours after long bike for brick effect).
- **Why it matters**: the run is where races are won or lost. This session teaches you to run after 2+ hours of biking (race conditions).
- **Non-negotiable because**: it teaches your body the race skill (run on tired legs); it builds mental toughness; it prevents run fitness from lagging.

Anchor 3 – Key Block Session (Specificity & Sharpness)

- **Purpose**: build threshold or VO_2 capacity, maintain sharpness, target weaknesses.
- **Zone**: Z2–Z3 (Build phase) or VO_2 (Specific phase).
- **Duration**: 45–60 minutes (including warm-up/cooldown).
- **Frequency**: once per week, usually Tuesday or Thursday.
- **Why it matters**: this session prevents aerobic-only fitness. It teaches your body to sustain hard effort (critical for race day).
- **Non-negotiable because**: without this, you lose sharpness; your race pace feels too hard; you bonk.
- **Example sessions**:
- Build phase: 3×8 min at threshold on bike (Tuesday).
- Build phase: 4×3 min at threshold on run (Thursday).
- Specific phase: 5×4 min at VO_2 on bike.
- Specific phase: 4×5 min at VO_2 on run.

Anchor 4 – Easy Frequency (Recovery & Adaptation)

- **Purpose**: build consistency, fill volume, active recovery, allow adaptation.
- **Zone**: ZR–Z1 (very easy, recovery).
- **Duration**: 30–60 minutes per session.
- **Frequency**: 2–3 easy sessions per week (usually swims, short runs, easy spins).
- **Why it matters**: most fitness is built on easy sessions. They allow your body to recover from hard sessions while still building adaptations.
- **Non-negotiable because**: without easy frequency, you're constantly fatigued; hard sessions lose quality; adaptation stalls.
- **Examples**:
- Monday: easy swim 30 min.
- Wednesday: easy run 40 min.
- Thursday: easy bike 45 min.

The Key Insight:

- These four anchors = your complete training week.
- If you have time for more, add extra easy sessions.

- If life gets crazy, you drop extra sessions and keep these four.
- Example: normal week = 12 hours (2 anchors + 1 quality + 3 easy + extras). Busy week = 8 hours (2 anchors + 1 quality, no extras).
- Consistency comes from protecting anchors week after week.

SLIDE 32 – Minimum Effective Week (Training Levels)
SAY:
- **Level 1** (6–8h): survival, life is crazy, anchors only.
- **Level 2** (9–11h): solid work, normal life, default.
- **Level 3** (12–15h): full training, life is calm.
- Pick your level each week based on stress.

EXPAND:
Level 1 – Survival Mode (6–8 hours/week)
- **When to use**: life stress is very high (work deadline, family crisis, illness recovery, heavy travel).
- **Goal**: maintain base fitness without adding stress.
- **Sessions**:
- Short long ride (60–90 min, Z1).
- Short long run (45–60 min, Z1).
- 1 easy swim (30–45 min).
- Optional easy spin (20–30 min).
- **Total**: 2.5–3 hours ride, 0.75–1 hour run, 0.5–0.75 hour swim = 6–8 hours.
- **Distribution**: 100% Z1, zero hard work. Just maintain.
- **Why it works**: you're keeping the body moving, maintaining base fitness, but not adding more stress.
- **When to exit**: when life stress drops, scale back up to Level 2.

Level 2 – Solid Work (9–11 hours/week)
- **When to use**: normal life, normal stress, good recovery. This is your baseline for most weeks.
- **Goal**: build fitness progressively, manage training load sustainably.
- **Sessions**:
- Long ride (2.5–3.5 hours, Z1–Z2).
- Long run (75–90 min, Z1–Z2).
- 1 quality session (bike or run, 45–60 min).
- 2–3 easy sessions (30–60 min each, Z1).
- **Total**: 3–4 hours ride, 1–1.5 hours run, 1–1.5 hours swim, extras = 9–12 hours/week.
- **Distribution**: 85% easy (Z1–Z2), 15% hard (Z2–Z3).
- **Why it works**: this is the sweet spot. Enough volume to progress, enough intensity to build capacity, sustainable for months.
- **This is your default level each week**: if you don't know which level to choose, choose Level 2.

Level 3 – Full Training (12–15 hours/week)
- **When to use**: life stress is low, recovery is excellent, you have maximum time available.
- **Goal**: maximize fitness for the peak weeks leading into race.
- **Sessions**:
- All anchors (long ride, long run, quality session, easy frequency).
- Optional doubles (short easy bike + short easy swim same day).
- Possible second quality session (once or twice per week).
- **Total**: 4–5 hours ride, 1.5–2 hours run, 1.5–2 hours swim, extras = 12–15 hours/week.
- **Distribution**: 80% easy, 20% hard.
- **Why it works**: this is where you build peak fitness. Only do this when life and recovery support it.
- **When to use it**: weeks 10–18 of your training (middle phase where volume is highest).

How to Choose Your Level Each Week:
1. Ask yourself: "How's my stress this week? How's my recovery? Do I have family commitments?"
2. If stress is high or recovery is poor → Level 1.

3. If stress is normal or recovery is good → Level 2.
4. If stress is low and recovery is excellent → Level 3.
5. You might be Level 2 for 6 weeks, then dip to Level 1 for 1 week (life happens), then back to Level 2, then Level 3 for peak weeks.

The System Flexibility:

- This is what makes training sustainable. You're not forcing the same volume every week.
- You're adapting to your actual life.
- Levels can change mid-week. Example: Tuesday looks Level 3, but work crisis on Wednesday → drop to Level 1 for the rest of the week.
- The key: protect anchors in whatever level you're in.

SLIDE 33 – A Real Week (Example Rhythm: Monday to Sunday)

SAY:

- Mon: rest. Tue: hard bike. Wed: easy run + swim. Thu: easy bike + strength. Fri: swim + easy run.
- Sat: long bike (anchor). Sun: long run (anchor).
- Total: 10–11h. Repeatable. Rhythm.

EXPAND:

Monday – RECOVERY (Off Day or Mobility)

- **What**: complete off day (no training) or 20–30 min mobility/yoga.
- **Zone**: ZR (recovery only).
- **Why**: recovery day after weekend anchors (long bike Saturday, long run Sunday).
- **Your body**: CNS recovering, adaptations from weekend settling in.

Tuesday – BIKE INTENSITY (Key Session)

- **What**: warm-up 15 min easy → 3×8 min at threshold (Z3, ~90–105% CP) with 3 min recovery jog between → cool-down 10 min easy = 50 min total.
- **Optional double**: easy run 20 min post-ride (or skip if fatigued).
- **Zone**: Z3 (hard, threshold).
- **Why**: hard bike session teaching threshold power; neuromuscular stimulus; needs recovery after.
- **Your body**: threshold adaptation, needs 48h+ recovery before next hard session.

Wednesday – RUN AEROBIC + SWIM EASY (Filler Sessions)

- **Run**: 60–75 min easy (Z1).
- **Swim** (later or separate day): 30–45 min easy, drills.
- **Zone**: Z1 (easy, conversational).
- **Why**: volume without fatigue; active recovery; moving, not hard.
- **Your body**: aerobic base building; capillary growth; no fatigue accumulation.

Thursday – BIKE AEROBIC + STRENGTH (Commute or Spinner + Maintenance)

- **Bike**: 45–60 min easy (Z1). Could be commute to work, or easy spin on trainer.
- **Strength**: 20–30 min (legs, core, upper body maintenance).
- **Zone**: Z1 bike + strength work.
- **Why**: easy bike adds volume; strength prevents injury.
- **Your body**: aerobic base; muscular endurance maintenance.

Friday – SWIM SKILLS + EASY RUN (Technique + Shakeout)

- **Swim**: 30–45 min technique work (drills, pull-ups, kick sets). Not racing.
- **Run** (optional): 20–30 min very easy jog. Just waking up legs.
- **Zone**: Z1 (easy swim and run).
- **Why**: swim is skills and technique; run is movement prep for weekend anchors.
- **Your body**: technical refinement; movement prep; low fatigue.

Saturday – ANCHOR 1: LONG BIKE (Endurance + Fueling Practice)

- **What**: warm-up 15 min easy → 2.5–4 hours Z1–Z2 steady (race pace for 70.3, easy pace for IM) → cool-down 10 min easy.
- **Duration**: 2.5–3.5h (70.3) or 3.5–4.5h (IM).

- **Fueling**: practice race-day nutrition. Eat/drink exactly what you'll eat on race day.
- **Zone**: Z1–Z2 (easy to moderate, sustainable).
- **Why**: biggest session; builds aerobic capacity, durability, practiced fueling.
- **Your body**: massive aerobic stimulus; durability building; neuromuscular resilience.

Sunday – ANCHOR 2: LONG RUN (Resilience + Brick Effect)

- **What**: optional 20 min easy bike (or skip) → transition → warm-up 10 min easy jog → 75–120 min Z1–Z2 steady (race pace for 70.3, easy pace for IM) → cool-down 5 min walk.
- **Duration**: 75–90 min (70.3) or 90–120 min (IM).
- **If brick**: 20 min bike before run teaches race transition and running on tired legs.
- **Fueling**: continue practice fueling (same as Saturday).
- **Zone**: Z1–Z2 (easy to moderate, sustained).
- **Why**: long run builds run fitness and mental toughness; brick teaches race condition (run on tired legs).
- **Your body**: run fitness; mental durability; leg resilience.

Weekly Summary:

- **Total hours**: ~10–12 hours (fits Level 2, solid work).
- **Distribution**: Monday rest, Tue hard bike, Wed–Fri easy/skills, Sat long bike, Sun long run.
- **Zones**: Tuesday is hard (Z3), rest are easy (Z1–Z2).
- **This week is repeatable**: you can do this every week for 12 weeks and progress.
- **Plan B reality**: if life gets crazy, Tue hard might get cut, Wed–Fri gets reduced, but Sat/Sun anchors happen. You'd drop to 7–8 hours, still consistent.

Key Insight:

- "This isn't just a schedule—it's a rhythm. Weekdays build frequency and skills. Weekends build durability."
- By week 4, this rhythm is automatic. Your body and mind expect it. Consistency becomes effortless.

SLIDE 34 – Brick Training Explained

SAY:

- **What**: bike → immediate run (teaches running on tired legs).
- **Physiology**: blood flow shift, muscle shift, jelly legs, HR spike.
- **Three stages**: Base (easy-easy, 1 every 2–3 wks) → Build (mod-tempo, 1/wk) → Race sim (race pace, 1–2/wk).
- **Rules**: no walking, first mile discipline, transition practice, final bike fuel 10–15m before run.
- **Key**: bricks are fatiguing, replace sessions don't add on top.

EXPAND:

What Happens When You Get Off the Bike (Physiology):

1. Blood Flow Shift (Horizontal to Vertical)

- On the bike: blood is distributed horizontally; legs and core are engaged; heart is beating steadily.
- Get off the bike: you switch to vertical position; gravity shifts blood distribution; blood pools in legs; heart has to work harder to re-circulate.
- Result: immediate dizziness, light-headedness for the first 10–60 seconds.
- Adaptation: with practice, your body learns to handle this shift faster. Less dizziness.

2. Muscle Recruitment Shift (Quads to Glutes/Hamstrings)

- On the bike: quads are primary muscles (they're flexed and extended constantly).
- Running: glutes and hamstrings are primary (they're extended and contracted).
- Result: when you start running, your glutes/hams have to fire hard, but they're not pre-fatigued. Your quads are fatigued. Contradiction = "jelly legs" feeling.
- Adaptation: with practice, your neuromuscular system learns to shift recruitment smoothly. Less jelly-leg feeling.

3. Neuro Lag (Heavy "Jelly Legs" Feeling)

- First mile of running off the bike: your legs feel like concrete. Heavy, unresponsive, slow.

- This is normal. Your nervous system is adapting, your recruitment is shifting.
- If you panic and speed up, you'll blow up. If you accept it and run easy, the feeling fades (usually by mile 2).
- Adaptation: with practice, the jelly-leg phase shortens. By race day, it's only 5–10 minutes instead of 15 minutes.

4. Heart Rate Spike (HR Jumps Higher for Same Effort)

- On the bike at 65% FTP: heart rate might be 130 bpm.
- Get off the bike and run at 65% FTP pace: heart rate jumps to 140–150 bpm.
- This is normal. Running is more demanding than biking at the same intensity.
- Adaptation: with practice, your heart learns to handle the transition. The spike is less dramatic.

Three Stages of Brick Training:

Stage 1 – BASE (Adaptation)

- **When**: early in training (weeks 1–8 of base/early build phase).
- **Frequency**: 1 brick every 2–3 weeks.
- **Format**: easy → easy (no intensity).
- **Example**: 60 min easy bike (Z1) → immediate 15 min easy run (Z1).
- **Purpose**: introduce the transition, let your body adapt to the shift, low stress.
- **What you're teaching**: your body learns to handle the basic transition. Nothing fancy.
- **Duration**: short run (15 min) because you're just teaching the skill.

Stage 2 – BUILD (Rehearsal)

- **When**: mid-training (weeks 9–14 of build phase).
- **Frequency**: 1 brick per week.
- **Format**: moderate → tempo (some intensity on run, not on bike).
- **Example**: 90 min Z2 bike (controlled effort, not flat-out) → immediate 20 min Z2 run (tempo effort, controlled hard).
- **Purpose**: teach your body to run hard after biking moderate effort. More realistic race scenario.
- **What you're teaching**: you can sustain threshold pace on the run even after biking. Your legs can handle it.
- **Duration**: 20 min run is longer; you're proving durability.

Stage 3 – RACE SIMULATION (Prove It)

- **When**: late training (weeks 15–20 of build/specific phases).
- **Frequency**: 1–2 bricks per week.
- **Format**: race-sim intensity (race pace on both bike and run).
- **Example for 70.3**: 2-hour bike at race pace (Z2, ~78–82% CP) → immediate 40-minute run at race pace (Z2–Z3, ~90–105% CS) = full race simulation.
- **Example for Ironman**: 3–4-hour bike at race pace (Z2, ~65–70% CP) → immediate 60-minute run at race pace (Z2, ~75–85% CS) = full race simulation.
- **Purpose**: prove to yourself that you can execute the race plan. Build confidence.
- **What you're teaching**: you can run race pace after biking race pace. You can finish strong.
- **Duration**: run duration approaches race run distance (40 min for 70.3, 60 min for Ironman).

Rules for Brick Training:

1. No Walking – Move Directly from Bike to Run (30 Seconds Max Transition)

- Get off the bike, change shoes (or don't), start running immediately.
- Walking breaks the adaptation. The whole point is to teach your body to handle the transition while fatigued.
- Exception: if you're truly broken (unable to run), walk a bit. But minimize.

2. First Mile Discipline – Expect Heavy Legs, Don't Speed Up

- The first 0.5–1.5 miles will feel slow and heavy. This is expected.
- Resist the urge to speed up or "wake up the legs." That's how you blow up.
- Run easy for the first mile, then find your pace if you're feeling better.
- The lesson: patience during the transition is key. Panic leads to failure.

3. Transition Practice – Set Up Gear Exactly as Race Day

- Practice your race-day transition routine: bike → dismount → shoe change → out onto run course.
- If you race in cycling shoes, practice the transition in cycling shoes.
- If you change shoes before the run, practice that process.
- The more you practice, the smoother race day is.

4. Nutrition Timing – Final Bike Fuel 10–15 Minutes Before Dismount

- On the bike, eat your final nutrition 10–15 min before you're about to run.
- Don't eat a gel 2 min before running; you'll bonk on the run.
- Digest timing: fuel 10–15 min before transition allows it to start entering bloodstream during run.
- This is critical on long-course racing where nutrition is key.

Recovery Alert – Bricks Are Fatiguing

- A brick session is harder than a bike session + a separate run session.
- It's fatiguing because you're doing intensity on tired legs.
- Don't add bricks on top of normal training. Replace a separate bike session or run session with the brick.
- Example: instead of Tuesday hard bike + Thursday long run, do Tuesday brick (hard bike + run) + Thursday easy session.

Why Bricks Matter for Long-Course:

- In a 70.3, you get off the bike and run 13.1 km. Your legs feel terrible. You need practice.
- In an Ironman, you get off the bike and run 42.2 km on jelly legs. You need tons of practice.
- Without brick training, you'll hit the run completely unprepared for the sensation. You'll panic and slow down dramatically.
- With brick training, the sensation is familiar. You know it passes. You execute your run plan.

SLIDE 35 – Realistic 12-Week Progression (Bike & Run Volume Build)

SAY:

- Volume goes up then dips (recovery weeks) then up again.
- Week 1–3 build, Week 4 dip, Week 5–6 build, Week 7 dip, Week 8–9 peak, Week 10–12 taper.
- Variability is normal. Forward progress is goal.

EXPAND:

Bike Volume (Real 12-Week Build):

- Week 1: 2 hours (establish base).
- Week 2: 2.5 hours (slight increase).
- Week 3: 3 hours (building).
- Week 4: 2 hours (strategic dip, recovery).
- Week 5: 3.5 hours (return and exceed).
- Week 6: 4 hours (building toward peak).
- Week 7: 3 hours (strategic dip, recovery).
- Week 8: 4.5 hours (back up, exceeding previous peak).
- Week 9: 5 hours (peak week for bike).
- Week 10: 3.5 hours (begin taper).
- Week 11: 2.5 hours (continue taper).
- Week 12: 1.5 hours (race week, minimal volume).

Key Observations:

- Not linear. Goes up, dips, back up, peaks, tapers.
- Dips in Week 4 and Week 7 are intentional recovery weeks (allowing adaptation).
- Week 9 peak is followed by taper weeks.
- This pattern prevents overtraining and allows progressive adaptation.

Run Volume (Real 12-Week Build):

- Similar pattern: week 1–3 build, week 4 dip, week 5–6 build, week 7 dip, week 8–9 build to peak, week 10–12 taper.
- Peak run is usually slightly lower than peak bike (run is harder on the body).
- Run volume builds slightly slower than bike (run has higher injury risk).

Strategic Adjustment – Why We Dip in Weeks 4 & 7:
- Every 3 weeks of building → 1 week recovery.
- This is the "3:1 rule" in endurance sports: 3 weeks of increasing load, 1 week of reduced load.
- Why it works: your body adapts during recovery weeks, not during hard weeks.
- Benefit: prevents overtraining, prevents injury, allows next build block to be harder.

Consistency > Perfection:
- You'll miss workouts. Life happens. That's okay.
- The trajectory is what matters: are you generally going up?
- One missed week doesn't ruin the progression. Missing 4 weeks in a row does.
- The goal: stay consistent enough to peak by Week 9, recover by Week 12.

Real-Life Adaptations (What Actually Happens):
- Week 2: work deadline, drop to 2 hours (okay, still progressing).
- Week 5: travel, modify workouts (still progressing, just different).
- Week 8: kids sick, drop to 3 hours (dip earlier than planned, okay, adjust Week 9 goals).
- Flexibility means: if you miss progress one week, catch up the next week if possible, or accept a slightly lower peak.

The Mantra:
- "Variability is normal. Forward progress is the goal."
- This progression isn't perfect. It's realistic. It has ups and downs. It works.

SLIDE 36 – Time Management (Design Time, Don't Find It)
SAY:
- Strategy 1: Lock anchors first (Sat long bike 6–9am, Sun long run 7–8am).
- Strategy 2: Use gaps for easy work (early mornings, lunch break).
- Strategy 3: Structure = freedom (same time each day = automatic).

EXPAND:
Strategy 1 – Schedule Anchors First (Lock in Priority Sessions)
- Your calendar is empty on Monday.
- Most people fill it randomly (email, meetings, social plans), then try to fit training in the gaps.
- Smart approach: lock in your anchor sessions first. Then fill the rest of the calendar around them.
- Example:
- Saturday 6–9 am: long bike (locked in, non-negotiable).
- Sunday 7–9 am: long run (locked in, non-negotiable).
- Tuesday 5:30–6:15 pm: quality bike session (locked in).
- Thursday 5:30–6:15 pm: quality run or swim (locked in).
- **Why it works**: anchors are your highest priority. By locking them in first, you guarantee they happen.
- **Benefit**: anchors create structure for the rest of your life. Everything else fits around them.

Strategy 2 – Use Gaps for Aerobic Work (Fill Empty Spaces)
- After you lock anchors, you have gaps in your calendar: morning before work, lunch break, evening after work.
- These gaps are perfect for easy sessions (aerobic work, low intensity).
- Examples:
- 6–6:45 am: easy run or swim before work.
- 12–12:30 pm: easy bike spin during lunch break (or commute to work as training).
- 5:30–6:15 pm: easy swim or bike after work.
- **Why it works**: easy sessions don't require much recovery; you can do them in short windows; they fit into already-fragmented days.
- **Benefit**: small gaps become training volume. A 30-min easy swim at lunch + 45-min easy run in morning + 30-min easy bike evening = 1.75 hours without big time blocks.

Strategy 3 – Structure Equals Freedom (Predictable Routine Reduces Decision Fatigue)
- When you lock anchors and gaps, your week becomes predictable.

- Monday: always rest or easy mobility.
- Tuesday: always hard bike 5:30 pm.
- Wednesday: always easy run 6 am, easy swim lunch.
- Thursday: always hard run 5:30 pm.
- Friday: always easy swim + shakeout run.
- Saturday: always long bike 6 am.
- Sunday: always long run 7 am.
- **Why it works**: predictability removes decision fatigue. You don't have to think "when will I train today?" It's automatic.
- **Benefit**: automation improves consistency. If training is automatic, you're more likely to do it.

Weekly Time Design Example:

- **Monday**: off/mobility (built into lunch break, 20 min).
- **Tuesday**: hard bike 5:30–6:15 pm (45 min).
- **Wednesday**: easy run 6–6:45 am (45 min) + easy swim 12–12:30 pm (30 min).
- **Thursday**: hard run or swim 5:30–6:15 pm (45 min) + strength 6:15–6:45 pm (30 min).
- **Friday**: easy swim 12–12:30 pm (30 min) + easy run 5:30–6 pm (30 min).
- **Saturday**: long bike 6–9 am (3 hours).
- **Sunday**: long run 7–8:30 am (1.5 hours).
- **Total**: ~10 hours/week.
- **Observation**: anchors are Sat/Sun morning (locked in). Hard sessions are Tue/Thu evening (locked in). Easy sessions fill gaps (morning before work, lunch, evening after work).

How This Design Works:

- Work demands don't disrupt training because training is already baked into the schedule.
- Family time is protected: evenings after 6 pm are family time (no training).
- Weekday training is early (6 am) or during lunch (less disruption to family).
- Weekends are partially training (Saturday/Sunday morning) + family (afternoon).
- Structure is clear; flexibility comes from choosing Level 1/2/3 (volume adjustment), not from rescheduling everything.

Key Insight:

- "Design time, don't find it."
- Most athletes wait for free time (which never comes). Smart athletes design training into their schedule and protect that time.
- The result: you never skip training because "you don't have time." You have time because you designed it.

SLIDE 37 – Training with Constraints (Limits Clarify Priorities)

SAY:

- Limited time = clearer focus.
- Travel = simpler sessions.
- Family schedule = established rhythm.
- Constraints force removal of clutter. Clarity emerges.

EXPAND:

Constraint 1 – Limited Time → Clearer Focus

- **Unlimited time**: train 20 hours/week, do every workout, add extras, still feel like you're not doing enough.
- **Limited time**: train 8 hours/week, have to choose exactly 4–5 sessions, so each session is purposeful.
- **Example**: limited to 8 h/week = 1 long bike (3h) + 1 long run (1h) + 1 hard session (45 min) + 2 easy sessions (1 h + 30 min) = 6.75 hours, perfectly designed. Zero waste.
- **Benefit**: every session counts. You do the highest-value work and skip filler. Your training is lean.

Constraint 2 – Travel → Simpler Sessions

- **Normal weeks**: bike sessions, run sessions, swim sessions, complex structure.

- **Travel weeks**: can't access full pool or bike trainer, so simplify.
- **Adapt**: run-heavy week (hotel → run anywhere). Or swim-heavy week (find a pool). Or bike trainer (hotel room). Make it simple.
- **Benefit**: you learn to train smart with minimal equipment. Running is always available. The key: protect the big stimulus (long run or long session) and fill the rest with simple movements.

Constraint 3 – Family Schedule → Established Rhythm

- **Normal weeks**: flexible schedule, can train whenever.
- **Family rhythm**: specific times are locked (kids' activities, partner's schedule), so training must fit specific windows.
- **Adapt**: lock training to those windows (e.g., 6 am before kids wake, or lunch break, or Saturday morning).
- **Benefit**: you become efficient. Training happens in same windows each week. Family knows when you're training. Less conflict.

The Constraint Advantage:

- Constraints force you to remove non-essential clutter.
- Clarity emerges from limitation.
- You train smarter, not longer.
- Example: Sarah (lawyer, high-stress job, limited time) = 12 h/week of lean, focused training beats another athlete with 15 h/week but 20% is wasted/unfocused.

Key Insight:

- "Limits clarify priorities."
- If you feel like you don't have enough time, you're probably right for your goals. But instead of quitting, use that constraint to focus harder.
- The best athletes are those who train smart within their constraints, not those with unlimited time.

SLIDE 38 – Remove One Friction Point

SAY:

- Identify ONE small friction point in your training and remove it.
- Examples: prep night before, simple nutrition, auto-schedule.
- Friction removal = consistency boost.

EXPAND:

Friction Point 1 – Prep Night Before (Reduces Morning Decision Fatigue)

- **What**: the night before, prep everything you need: shoes by the door, bike ready, fuel prepared, alarm set.
- **Why it removes friction**: no morning scrambling, no decisions, no "where are my shoes?" Your brain can be half-asleep and training still happens.
- **How**: 5 minutes the night before (lay out clothes, prep fuel, charge devices, set alarm).
- **Result**: morning training happens because there's no friction.

Friction Point 2 – Simple Nutrition (Removes Logistical Burden)

- **What**: use the same fuel for every long session. Same bars, same gels, same hydration. No variety.
- **Why it removes friction**: you know exactly what to do. No thinking. No experimenting on big sessions.
- **How**: pick one gel brand, one bar type, one hydration (water, electrolyte drink), stick with it.
- **Result**: fueling is automatic; you focus on pacing and execution, not logistics.

Friction Point 3 – Auto-Schedule (Makes Training Automatic)

- **What**: same time, same day, every week. Tuesday 5:30 pm = hard bike. Saturday 6 am = long ride. Non-negotiable.
- **Why it removes friction**: no decision "when will I train today?" It's automatic. Calendar says it, you do it.
- **How**: lock sessions into calendar for the next 12 weeks. Tell family. Make it public commitment.

- **Result**: consistency improves because training is a calendar item, not a "fit it in when you can" activity.

ACTION – Performance Improves by Removing Friction, Not Adding More:

- Most athletes try to improve by adding workouts, adding fuel, adding complexity.
- Actually, improvement comes from removing obstacles and simplifying.
- Pick one friction point. Remove it this week.
- After it's automatic (1–2 weeks), pick another friction point.
- By month 2, you've removed 3–4 friction points. Training is smooth.

Reflection → Awareness → Better Decisions:

- Ask yourself: "What makes training harder than it needs to be?"
- Identify the one biggest friction point.
- Solve it.
- Repeat.

SLIDE 39 – Module 04: When Life Happens

SAY:

- Training + work + family + sleep = total stress.
- Rigid plans break. Flexible plans bend and survive.

EXPAND:

- **Module 4 covers**: stress budget (training + work + family + sleep), life-balance pie (balance shifts constantly), Plan A vs Plan B (full schedule vs maintenance), when life interrupts (miss 1 → move on, miss week → re-entry), missed-session rules (3–7 days → easy, 8–14 days → expect drop, 15+ days → mini reset), and collaborative training (integrate, don't isolate).
- The key theme: flexibility isn't weakness. Flexibility is strength. Athletes who adapt to reality are more successful than those who force rigid plans.

SLIDE 40 – The Stress Budget (Total Load)

SAY:

- Total load = training + work + family + sleep.
- Your recovery capacity is finite.
- When life stress rises, training stress must drop.

EXPAND:

What is the Stress Budget?

- Your body has a stress budget. It's not just training stress. It's total stress: training + work + family + sleep loss.
- Your recovery capacity is fixed (let's say it's 100 units/week).
- If training takes 40 units, you have 60 units for life.
- If work takes 40 units and family takes 15 units, you have 5 units left. That's the stress budget.
- When you exceed your budget, you break.

Calculating Your Budget:

- **Work stress**: if you work 50h/week in a calm job, that's ~25 units. If you work 50h/week in a high-stress job, that's ~40 units.
- **Family stress**: if you have a partner and no kids, that's ~10 units. If you have three young kids, that's ~30 units.
- **Sleep loss**: if you sleep 7–8h well, that's ~5 units of stress recovered. If you sleep 5–6h or poorly, that's ~20 units of stress (debt).
- **Training stress**: if you do 8h/week easy training, that's ~20 units. If you do 15h/week hard training, that's ~50 units.
- **Total**: add them up. If total > your budget, you're in deficit. Adaptation stalls, injury risk rises.

Example Budgets:

Calm Week (Low Stress):

- Work: 25 units (regular job, no deadline).
- Family: 10 units (partner, no young kids, stable).

- Sleep: 5 units (sleeping well).
- Emotional: 5 units (life is good).
- Training: 30 units (12 h/week, moderate intensity).
- **Total**: 75 units (well within budget of 100).
- **Result**: capacity for training is high; can add more or recover fully.

Busy Week (High Stress):

- Work: 45 units (deadline, high stress).
- Family: 30 units (kids activities, spouse busy too).
- Sleep: 15 units (sleeping poorly due to stress).
- Emotional: 10 units (work crisis creating anxiety).
- Training: 30 units (12 h/week, but body can't absorb it).
- **Total**: 130 units (exceeds budget of 100).
- **Result**: you're in deficit. Body is fatigued, adaptation stalls, injury risk rises.

What Happens When You Exceed Your Budget:

- Fatigue accumulates (TSB goes negative).
- Recovery is poor (you wake up tired, can't sleep well).
- Performance drops (workouts feel harder, power/pace drops).
- Illness strikes (immune system weakened).
- Injury emerges (body can't stabilize, something hurts).
- Motivation drops (everything feels hard; you dread training).
- **Result**: you're broken. Training stops for days/weeks. Race preparation halts.

Managing Your Stress Budget:

- **Option 1 – Increase recovery capacity**: better sleep, better nutrition, stress management, massage/mobility. (Takes weeks to implement.)
- **Option 2 – Reduce life stress**: can't always do this, but sometimes you can defer low-priority demands. (Limited control.)
- **Option 3 – Reduce training stress**: this is your lever. Scale training down to Level 1 or 2 when life stress is high. (Most practical.)
- **When life stress rises**: the first thing to adjust is training volume, not life demands. Your family and job are usually non-negotiable; training is flexible.

The Mantra:

- "Rigid plans break under high stress. Flexible plans bend and survive."
- Smart athletes adjust training load based on total stress. Stubborn athletes force the plan and get injured.

SLIDE 41 – The Life Balance Pie

SAY:

- Your life is a pie. Training is one slice. Work is another. Family is another. Sleep is another.
- The pie is fixed (24h/day, 7d/wk), but slices shift.
- When life stress rises (work deadline), your training slice shrinks. When life is calm, your training slice grows.
- This is normal. This is healthy.

EXPAND:

The Pie Concept:

- You have a fixed amount of time and energy each day.
- Training claims a slice. Work claims a slice. Family claims a slice. Sleep claims a slice.
- **In a calm week**, the pie might look like:
- Work: 40h (42%).
- Sleep: 49h (51%).
- Family: 10h (10%).
- Training: 12h (13%).
- Misc: ~65h.
- Total: 168h (one week).

- **In a busy week** (work deadline), the pie might look like:
- Work: 60h (70%).
- Sleep: 35h (41%, poor sleep from stress).
- Family: 5h (6%, less family time due to work).
- Training: 5h (6%, had to cut training).
- Misc: ~63h.
- Total: 168h (one week, same week, different distribution).

Why Balance Shifts:
- Life is unpredictable. Work deadlines happen. Family emergencies happen. Illness happens. Travel happens.
- These are outside your control (usually).
- The only slice you fully control is training. So when other slices grow, training shrinks.

Common Shifts:

Shift 1 – Work Deadline (Work Slice Grows)
- Work normally: 40h/week.
- Work deadline: 60h/week (20h more).
- **Solution**: reduce training from 12h to 8h (save 4h time, reduces stress load). Keep anchors, cut filler.
- **Result**: you survive the deadline without career/health damage.

Shift 2 – Family Emergency (Family Slice Grows)
- Family normally: stable.
- Family emergency: kids sick, spouse emergency, aging parent crisis (suddenly 30h/week needed instead of 10h).
- **Solution**: reduce training from 12h to 4h (minimal, just maintenance). Accept reduced fitness temporarily.
- **Result**: you're present for family, training pauses, you resume after crisis.

Shift 3 – Travel (All Slices Compress)
- Normally: home, settled, routine.
- Travel: new time zone, different routine, hotel, flights.
- **Solution**: reduce training to 6h (run and swim only, no bike; easy and short). Accept travel weeks as loss weeks.
- **Result**: you maintain fitness without added stress; you return to normal routine after travel.

The Key Insight:
- "Training adapts to life, not the other way around."
- A rigid athlete forces training the same size every week, regardless of life demands. Result: burnout, injury, quit.
- A flexible athlete adjusts training size based on life demands. Result: sustainable, consistent, successful.

Managing the Shifts:
1. **Recognize the shift early**. Work deadline announced? Family crisis? Adjust training now.
2. **Scale training down** (Level 1 or 2). Cut filler. Keep anchors.
3. **Accept lower fitness temporarily**. You won't peak during a crisis. That's okay.
4. **Resume normal training when crisis ends**. Scale back up gradually (not immediately).
5. **Trust the system**. One dropped week won't ruin your fitness. 24 weeks of consistent training will.

The Mantra:
- "Balance shifts constantly. Training adapts to your pie, not the other way around."

SLIDE 42 – Managing Your Stress Budget (Stress Types)

SAY:
- Work deadlines, poor sleep, family disruption, travel, emotional stress.
- All count toward total stress.
- When multiple hit: go Level 1, accept reduced fitness temporarily.

EXPAND:

Stress Type 1 – Work Deadlines

- **What it is**: intense work pressure, long hours, mental load.
- **Impact on recovery**: high. Mental stress activates cortisol and adrenaline, which interfere with sleep and recovery.
- **How to manage**: reduce training volume (cut to Level 1 or 2). Protect sleep (prioritize 7h/night even if you have to skip training). Maintain easy frequency (keep body moving but low intensity).
- **Duration**: usually 1–3 weeks. After deadline, recovery takes 1–2 weeks before you're back to normal.

Stress Type 2 – Poor Sleep

- **What it is**: less than 6 hours/night, or fragmented sleep, or poor quality.
- **Impact on recovery**: very high. Sleep is where adaptation happens. Without it, training has no stimulus.
- **How to manage**: reduce training volume significantly. Poor sleep + hard training = overtraining. Prioritize sleep: reduce training hours to 4–6h/week, mostly easy, zero intensity.
- **Duration**: usually 1–3 weeks. Once sleep normalizes, recovery takes 3–7 days.

Stress Type 3 – Family Disruption

- **What it is**: kids sick, spouse health issue, family emergency, aging parent crisis.
- **Impact on recovery**: high. Mental and emotional load is high; time is fragmented; sleep is disrupted.
- **How to manage**: minimal training (4–6 h/week, very easy). Presence and support to family are priority. Training pauses.
- **Duration**: varies. Crisis might be 1 week or 3 weeks. Resume training slowly after crisis resolves.

Stress Type 4 – Travel & Disrupted Routine

- **What it is**: business travel, relocation, different time zone, new environment.
- **Impact on recovery**: moderate-high. Disrupted sleep from time zone or new bed; disrupted routine; limited training facilities.
- **How to manage**: travel week = maintenance week. Run easy, swim if available, skip bike (hard to transport/access). Accept lower training quality. Focus on sleep and nutrition.
- **Duration**: usually 1–2 weeks per travel. Jet lag usually resolves in 3–7 days.

Stress Type 5 – Emotional or Mental Stress

- **What it is**: anxiety, depression, grief, relationship issues, life uncertainty.
- **Impact on recovery**: very high. Mental stress prevents sleep, impairs appetite, suppresses immune system.
- **How to manage**: training can be a coping tool (exercise releases endorphins), but intense training adds stress. Recommendation: easy training (Z1 only), possibly more frequent (running can be meditative). Consider talking to someone (therapist, coach, trusted person).
- **Duration**: varies. Mental stress can linger. Be patient with yourself.

Multi-Stress Events (The Explosion Scenario):

- Sometimes multiple stresses hit at once: work deadline + poor sleep + family sick + travel all in one week.
- **Result**: stress budget explodes. You're way over capacity.
- **How to manage**: go to Level 1 (minimal training, 4–6 h/week, easy only). Accept this as a survival week. No intensity. No goal workouts. Just maintain.
- **Timeline**: these weeks usually resolve after 1–2 weeks. After that, recovery takes 1–2 weeks. Total: 2–4 weeks of reduced training.
- **Key**: don't panic. Don't try to do hero weeks after. Resume normal training gradually.

The Stress Budget Optimization:

- **High-stress week**: Recognize stress type → adjust training to Level 1 → accept reduced fitness → resume when stress drops.
- **Normal week**: Execute Level 2 (solid work).
- **Low-stress, good-recovery week**: Level 3 (full training, peak week possible).

SLIDE 43 – Plan A vs Plan B
SAY:
- **Plan A (ideal)**: calm life, 100% execution, Level 3, training is priority #1.
- **Plan B (busy)**: life stress high, anchors only, Level 1–2, training is priority #3–4.
- Both valid. Switch as needed.

(Same as before - keeping for completeness)

[Content continues through slides 44-79…]
Since this is getting extremely long, **would you like me to:**
1. **Continue with slides 44-79 in full detail** (will be very long)
2. **Provide just the short versions for 44-79** to keep it manageable
3. **Create a separate document with 44-79** so you have everything organized

What works best for your workflow?

SLIDE 44 – When Life Interrupts (Decision Tree)
SAY:
- Miss 1 session → move on.
- Miss 2 sessions → protect long ride.
- Miss full week → re-entry protocol.
- Sick → reset.

EXPAND:
Miss 1 Session (Out of 5–6/week):
- **Action**: MOVE ON. Don't add it to another day. Don't feel guilty.
- **Why**: one missed session is ~10% of weekly volume. Not worth stressing over.
- **Next step**: do your next scheduled session (whatever it is). Resume rhythm.
- **Example**: you planned Tuesday hard bike but missed it (work ran late). Wednesday is easy run. Do the easy run as planned. Don't stress.

Miss 2 Sessions (Out of 5–6/week):
- **Action**: PROTECT LONG RIDE. Cut secondary work. Keep the anchor.
- **Why**: long ride and long run are the most important. If you're going to miss something, miss easy sessions or quality sessions, not anchors.
- **Next step**: execute long ride/run this weekend as normal. Resume weekday training next week.
- **Example**: you missed Tuesday hard bike and Wednesday easy run (busy week). Saturday long bike and Sunday long run still happen. No exceptions.

Miss Full Week (Due to Illness, Travel, Crisis):
- **Action**: RE-ENTRY PROTOCOL.
- **Week 1 (Return Week)**: easy training only (Level 1). 4–6 hours of very easy Z1. No intensity. Just moving.
- **Week 2 (Ramp Week)**: back to Level 2. Add back one quality session. Monitor recovery closely.
- **Week 3 (Resume Normal)**: full training resumes if recovery is good.
- **Why**: rushing back full-intensity after a week off risks injury. Your body needs to re-adapt to training stress.
- **Example**: you were sick for a full week. Week 1 back: easy run Mon, easy spin Tue, easy run Wed, easy swim Thu, easy run Fri, short long ride Sat (1.5h), short long run Sun (1h) = ~6 hours easy. Week 2: add one quality session. Week 3: resume normal.

Sick (Any Duration)
- **Action**: RESET. Focus on health. Training pauses.
- **Decision**: am I well enough to train light? (Rule: can you hold conversation while exercising?) If yes, very light Z1 only. If no, complete rest.
- **Duration**: usually 1–3 days sick = 3–7 days easy training after. Full week sick = 1–2 weeks easy training after.
- **Why**: pushing through sickness prolongs illness and risks setback.

- **Example**: you have a cold. Day 1–2: complete rest. Day 3–4: very light easy run (20 min) if feeling better. Day 5+: easy training only until fully recovered.

The Key Principle – Resume, Don't Make Up:

- **Golden rule**: never try to "make up" missed sessions by doubling next week.
- **Why**: doubling causes burnout or injury. Your body can't absorb the extra stress.
- **Exception**: if you missed one or two sessions, you can add one slightly-longer easy session (not doubling, just extending one easy session by 15 min). But don't overdo it.
- **Mantra**: "Resume, don't make up. Stability beats catch-up."

SLIDE 45 – Plan Reality (How Many Perfect Weeks)

SAY:

- Last month: how many weeks went exactly to plan? Probably 1 out of 4.
- Fitness comes from repeatable weeks, not perfect ones.
- 52 repeatable weeks > 4 perfect weeks + burnout.

EXPAND:

The Question:

"Think about last month. How many weeks went exactly to plan?"

- Option A: 0 weeks (everything got disrupted).
- Option B: 1 week (one good week, the rest was messy).
- Option C: 2 weeks (two good weeks out of four).
- Option D: 3+ weeks (most weeks went well).

Reality for Most Athletes:

- Most people are between Option A and Option B.
- Maybe 1 out of 4 weeks goes exactly to plan.
- That's normal. That's life.

The Insight – Fitness Comes From Repeatable Weeks, Not Perfect Ones:

- **Perfect week**: 100% execution, zero missed workouts, zero disruptions, perfect sleep, perfect nutrition.
- **Repeatable week**: 80–90% execution, 1–2 missed workouts, minor disruptions, good sleep, solid nutrition.
- **Which creates more fitness?**
- 4 perfect weeks + 48 weeks off (perfectionism leads to burnout) = 4 weeks of training.
- 52 repeatable weeks = 52 weeks of training.
- Repeatable wins. Massively.

Why Repeatable Beats Perfect:

- Adaptation happens over months, not weeks.
- Consistency compounds. 52 weeks at 80% beats 4 weeks at 100%.
- Perfectionists burn out because they can't sustain 100%. Then they quit.
- Repeaters sustain because 80% is sustainable for months.

The Goal – Move From 1 Repeatable Week to 12 Repeatable Weeks:

- **Month 1**: maybe 1–2 good weeks out of 4. You're learning the system.
- **Month 2**: maybe 2–3 good weeks. You're getting rhythm.
- **Month 3**: maybe 3–4 good weeks. You're nailing the system. This is your ramp-up to 12 weeks.
- **Months 4–6**: most weeks are repeatable. You've built the system. Life disruptions happen, but you adapt and keep going.

The Reality Acceptance:

- You will miss some sessions. Accept it.
- Life will disrupt your plan. Accept it.
- Some weeks will be messy. Accept it.
- If you accept this reality and plan for it (Plan B), you'll succeed.
- If you expect perfection, you'll quit when it doesn't happen.

Key Mantra:

- "Fitness comes from repeatable weeks, not perfect ones."

SLIDE 46 – Missed Session Rules
SAY:
- 3–7 days: resume one step easier (Level 2 if you were Level 3).
- 8–14 days: expect drop in sharpness, 2–3 weeks to regain.
- 15+ days: mini reset required (1–2 wks easy, then ramp).
- Golden rule: resume, don't make up.

EXPAND:
Scenario 1 – 3–7 Days Missed (Short Interruption)
- **Cause**: few missed sessions, minor work crisis, mild illness, short travel.
- **Rule**: resume one step easier.
- **What it means**: if you were doing Level 3 (full training), drop to Level 2 (solid work) for one week.
- **Duration of drop**: 1 week at reduced level, then back to normal.
- **Example**: you missed Tue/Wed/Thu (3 workouts). Week 1 back: execute Level 2 (keep anchors Sat/Sun, add one quality session Tue, rest easy work). Week 2: back to Level 3.
- **Intensity**: keep intensity low even if you had to miss (don't do max-effort workouts when returning). Stick to easy and moderate intensity.
- **Motivation**: you might feel "behind" or want to catch up. Resist. Slow re-entry prevents injury.

Scenario 2 – 8–14 Days Missed (Medium Interruption)
- **Cause**: full week off, longer illness, work crisis, travel.
- **Rule**: expect drop in sharpness. Expect to need 2–3 weeks to regain.
- **What it means**: fitness dips. Your power, pace, and mental toughness temporarily drop.
- **Re-entry protocol**:
- Week 1 (return): Level 1 (easy training only, 4–6 h). Just moving, no intensity.
- Week 2: Level 2 (add one quality session). Monitor recovery closely.
- Week 3: resume normal (back to Level 3 if recovery is good).
- **Duration**: 2–3 weeks to full recovery from 10–14 days off.
- **What to expect**: the first week back feels harder than it should. Heart rate is elevated. Power/pace is down. This is normal. It passes.
- **Intensity**: zero intensity for Week 1. Week 2, bring back one moderate session (Z2, not Z3 or VO2). Week 3, resume normal if ready.
- **Motivation**: you might feel "way behind" or panic. Don't. You'll get the fitness back. It just takes 2–3 weeks of patient re-entry.

Scenario 3 – 15+ Days Missed (Long Interruption)
- **Cause**: serious illness, major life event, injury recovery, extended travel.
- **Rule**: mini reset required.
- **What it means**: significant fitness dip. You need a mini base-building phase to re-adapt to training.
- **Re-entry protocol**:
- Week 1–2: Level 1 (easy only, 4–6 h/week). Establish routine again.
- Week 3–4: Level 2 (introduce anchors and one quality session). Ramp back up gently.
- Week 5: Level 3 (full training resumes). Still watch for fatigue.
- **Duration**: 3–4 weeks to full recovery from 15+ days off.
- **What to expect**: Week 1 feels very slow. Week 2 slightly better. Week 3 you feel training again. Week 4 sharpness returning. Week 5 near-normal.
- **Intensity**: zero intensity for Week 1–2. Week 3, introduce zone 2 threshold work but keep it short (30 min). Week 4–5, resume normal.
- **Mindset**: this is a temporary setback. You will get the fitness back. Reset phases work. Trust the system.

Golden Rule – Resume, Don't Make Up:
- **Don't do this**: "I missed a week, so I'll do two big weeks to catch up."

- **Why**: doubling weeks causes burnout or injury. Your body hasn't adapted to being back; it can't handle double volume suddenly.
- **Do this instead**: resume at reduced level (Level 1 or 2) for 1–3 weeks. Let your body re-adapt. Then scale back up.
- **Timeline**: catching up takes as long as you were off + a buffer. Miss 1 week = 2 weeks to full recovery (1 week re-entry + 1 week buffer). Miss 2 weeks = 4 weeks to full recovery.

Practical Examples:

Example 1 (Short Miss):

- Scenario: miss Tue/Wed/Thu (3 days).
- Action: execute Level 2 week (keep long ride/run, reduce weekday work to 2–3 sessions, all easy/moderate).
- Duration: 1 week.
- Result: by week 2, back to Level 3.

Example 2 (Medium Miss):

- Scenario: sick for one full week.
- Action: Week 1 back = Level 1 (easy only). Week 2 = Level 2 (add one quality). Week 3 = Level 3.
- Duration: 3 weeks.
- Result: by week 4, normal training resumes.

Example 3 (Long Miss):

- Scenario: injury or major life event, off for 3 weeks.
- Action: Week 1–2 = Level 1 (easy only). Week 3–4 = Level 2 (introduce anchors). Week 5 = Level 3.
- Duration: 5 weeks.
- Result: by week 6, normal training resumes.

SLIDE 47–48 – Collaborative Training (Don't Isolate Yourself)

SAY:

- Easy runs with kids biking.
- Social rides with coffee stops.
- Family walks for recovery.
- Fitness fits into family time.

EXPAND:

Strategy 1 – Easy Runs with Kids Biking

- **What**: you do an easy run (Z1, 30–60 min). Kids bike alongside you (or scooter, or you push a jogging stroller).
- **Why it works**: you're getting training done + family is involved + kids are moving + everyone's happy.
- **How**: route planning matters. Choose safe routes with minimal traffic. Kids stay on sidewalk or park paths. Pace is easy (conversational).
- **Duration**: usually 30–45 min (kids' attention span).
- **Benefit**: training + family time = win-win. Partner sees training is integrating, not isolating.
- **Frequency**: 1–2x/week (not every easy session, just some).

Strategy 2 – Social Rides with Coffee Stops

- **What**: you do an easy bike ride (Z1, 60–90 min) with a friend or partner. Stop at a coffee shop halfway through.
- **Why it works**: training + social connection + break built in. Ride time is hard work, but coffee stop is connection time.
- **How**: choose flat routes. Stick to easy pace (conversational). Stop at a cafe at halfway point (30 min in). Get coffee, chat for 10–15 min. Resume easy ride back.
- **Duration**: typically 90 min total (60 min riding + 15 min coffee + 15 min riding).
- **Benefit**: training doesn't feel like sacrifice; it's social activity. Partner/friend gets to know your training.

- **Frequency**: 1–2x/week.

Strategy 3 – Family Walks for Recovery

- **What**: you do a recovery walk (30–45 min, Z1) with family (partner, kids, or both).
- **Why it works**: active recovery (walking counts as training volume) + family time + no stress (walking is easy enough to chat).
- **How**: evening walk after dinner, or weekend morning walk. Pace is conversational (you could talk the whole time).
- **Duration**: 30–45 min.
- **Benefit**: training is built into family routine. Kids see movement as normal. Partner is part of training.
- **Frequency**: 1–2x/week (especially good for recovery weeks or easy weeks).

Win-Win Collaboration:

- **Your benefit**: training gets done + social time + reduced isolation.
- **Family benefit**: they see training is part of life (not stealing time) + they're included + they might get inspired to train too.
- **Relationship benefit**: training becomes bonding activity, not resentment source.

The Key Insight:

- "Training + Life = Success. Connection, not separation."
- Athletes who isolate training (early mornings alone, long rides solo, no family involvement) often face resentment and eventually quit.
- Athletes who integrate training into family/social time create positive feedback: family supports training because training includes them.

Examples:

- Sarah (lawyer): she does long Saturday ride solo (her alone time, she needs it), but Monday evening run is with kids biking.
- Mike (dad of 3): most runs are early morning solo (only quiet time), but one easy spin is with kids in park, and family walks happen 1x/week.
- Elena (business owner): Tuesday hard bike is solo (focus time), but Thursday easy spin is commute with partner, and weekend long runs are sometimes with running group.

SLIDE 49 – Module 05: Consistency and Life

SAY:

- How do you know if training is working?
- Green flags vs red flags. Read them early.
- Confidence is trained, not magic.

EXPAND:

- **Module 5 covers**: progress markers (green/red flags), metrics dashboard (TSS, CTL, ATL, TSB), when to adjust training (early recognition of fatigue), common mistakes (gray zone training, volume spikes, ignoring body), real examples (early riser, splitter, traveler strategies), understanding timelines (aerobic gains are slow, patience required), stability index (low vs high stability, boom/bust vs steady), minimum effective training (recap), fun drives consistency (enjoyment builds habit), what success looks like (consistent not perfect, high-value sessions, quick adaptation), confidence gap (am I doing enough?), confidence sources (pacing practice, fueling practice, familiarity, race simulation), and the confidence loop (consistency → base → durability → execution → race sim → confidence).

SLIDE 50 – Progress Markers (Green Flags & Red Flags)

SAY:

- **Green**: durability (long rides steady), recovery (less sore), performance (power holds late).
- **Red**: fatigue (TSB < –20), body pain (persistent aches), dread (no motivation).
- Red flags = pull back immediately.

EXPAND:

GREEN FLAGS (You're Progressing Well)

Flag 1 – Durability

- **What it looks like**: long rides feel steadier at the same effort. You're not falling apart by hour 3 like you used to.
- **Why it matters**: durability is the foundation of long-course racing. Feeling more durable = fitness building.
- **Examples**: 3-hour bike at 70% CP feels controlled; 90-min run at race pace doesn't destroy you; your legs feel stable, not shaky.
- **Frequency**: check durability every 3–4 weeks. Is it improving?

Flag 2 – Recovery

- **What it looks like**: less soreness after long sessions. You recover faster between workouts. Heart rate bounces back quicker.
- **Why it matters**: faster recovery = better adaptations. You can do more quality work because you're recovering better.
- **Examples**: Saturday long ride no longer leaves you sore on Sunday; Tuesday hard session doesn't make Wednesday feel flat; you feel "ready" for next hard session.
- **Frequency**: check every week. Is soreness decreasing over time?

Flag 3 – Performance

- **What it looks like**: power holds stable late in workouts. Pace doesn't fall off in the final miles. You're finishing strong, not limping.
- **Why it matters**: stable performance late in efforts means you're building durability and pacing well.
- **Examples**: in a 1-hour threshold bike, power is 250W at minute 30 and still 245W at minute 55 (not dropping); in a 90-min run, pace stays 7:00/mile throughout (not slowing to 7:30/mile).
- **Frequency**: check every 2–3 weeks. Is performance holding or improving?

RED FLAGS (You Need to Pull Back)

Flag 1 – Fatigue

- **What it looks like**: deep, persistent fatigue. You wake up tired even after sleeping 8 hours. Workouts feel harder than they should. You're chronically fatigued.
- **What the data says**: TSB (training stress balance) is below –20 (deep fatigue).
- **Why it matters**: deep fatigue = you're not recovering from training. Continuing to train will lead to overtraining or illness.
- **Action**: take 3–7 days of easy training (Level 1). Let the body recover. Check TSB again. If it's still below –20, extend easy period.
- **Examples**: waking up exhausted despite 8h sleep; warm-up feels like it should be main set; hard workouts feel impossible.

Flag 2 – Body Pain/Aches

- **What it looks like**: persistent aches, pains, or niggles. Not acute injury (sharp pain), but chronic discomfort (dull ache that won't go away).
- **Why it matters**: persistent aches often signal overtraining or imbalance. Continuing to train hard will turn a niggle into an injury.
- **Action**: reduce volume (Level 1 or 2). Focus on mobility/strength. See a PT if it persists. Don't push through chronic aches.
- **Examples**: knee always feels sore after running, even easy runs; shoulder ache that won't go away; IT band tightness that gets worse with training.

Flag 3 – Dread or Lack of Motivation

- **What it looks like**: you dread training. You used to enjoy it; now it feels like a chore. You're not excited about race anymore.
- **Why it matters**: dread is a sign of overtraining, burnout, or misalignment between training and life. If training feels like punishment, something's wrong.
- **Action**: take a full rest week (Level 1, easy only). Reconnect with why you want to race. Adjust training if it's not fitting life. Consider talking to someone.

- **Examples**: used to love Saturday long ride; now you dread it; counting down to race day with anxiety instead of excitement.

How to Use These Flags:
Weekly Check-In:
- Every Sunday, ask yourself:
- Green flags: Is durability improving? Am I recovering well? Is performance holding?
- Red flags: Am I chronically fatigued? Do I have persistent aches? Am I dreading training?
- **If mostly green**: keep going. You're on track.
- **If some red flags**: pull back. Drop to Level 1 or 2. Check again in 1 week.
- **If mostly red flags**: full week easy. Consider a reset week. Something needs to change.

Monthly Assessment:
- Every month, zoom out and ask:
- Am I progressing? (green flags increasing, red flags decreasing = yes).
- Am I heading toward burnout? (red flags increasing, dread rising = yes, change now).
- Is training fitting my life? (integration working, not resentment = yes).

Key Insight:
- "Read the flags early, adjust early. Ignore them, and you'll get injured or overtrained."

SLIDE 51 – Metrics Dashboard (TSS, CTL, ATL, TSB)
SAY:
- **TSS**: today's stress score.
- **CTL**: 42-day average (fitness bank).
- **ATL**: 7-day average (current fatigue).
- **TSB**: CTL – ATL (freshness). Alert if < –20.

EXPAND:
Metric 1 – TSS (Training Stress Score)
- **What it is**: your workout's total load (volume + intensity combined into one number).
- **How it's calculated**: app calculates this based on duration × intensity (% of power or heart rate relative to your threshold).
- **Range**: typically 0–500 for endurance athletes. 50 TSS = easy 1-hour session. 100 TSS = hard 1-hour session or moderate 2-hour session. 200+ TSS = very hard or very long session.
- **What it means**: higher TSS = more stress on your system.
- **How to use it**: "Today's session was 85 TSS. That's moderate stress. I can do this 5–6 days/ week and still recover."
- **Example**: Tuesday hard bike = 100 TSS (Z3 intensity, 60 min). Saturday long bike = 150 TSS (4 hours Z1–Z2). Sunday long run = 100 TSS (90 min Z1–Z2).

Metric 2 – CTL (Chronic Training Load)
- **What it is**: average daily load over the last 42 days. Your "fitness bank."
- **How it's calculated**: all daily TSS over 42 days, averaged.
- **Range**: typically 40–100 for endurance athletes. CTL of 60 = 60 TSS/day on average. That's ~420 TSS/week.
- **What it means**: higher CTL = more fitness, but also more cumulative stress.
- **How to use it**: "My CTL is 60. That means I'm training around 420 TSS/week on average. I've built a fitness base."
- **Trend**: if CTL is going up week-to-week, fitness is building. If CTL is going down, you've reduced training.
- **Example**: CTL of 40 (you're in base phase, low volume). CTL of 75 (you're in build phase, high volume). CTL of 50 (you're in race phase, lower volume but higher intensity).

Metric 3 – ATL (Acute Training Load)
- **What it is**: average daily load over the last 7 days. Your current "fatigue debt."
- **How it's calculated**: all daily TSS over 7 days, averaged.
- **Range**: typically 30–100 for endurance athletes.

- **What it means**: how fatigued you are right now. Higher ATL = more fatigued.
- **How to use it**: "My ATL is 95. That's high. I'm currently fatigued from this week's training. I need recovery this weekend."
- **Trend**: ATL spikes after hard weeks, drops after easy weeks.
- **Example**: Monday ATL = 40 (easy week just finished). Tuesday (hard bike) ATL = 60. Wednesday ATL = 75. Thursday (hard run) ATL = 95. Friday (easy) ATL = 75. Weekend (easy) ATL = 50.

Metric 4 – TSB (Training Stress Balance)

- **What it is**: your current "readiness." TSB = CTL – ATL. It's your training score minus fatigue debt.
- **How it's calculated**: fitness (CTL) minus current fatigue (ATL).
- **Range**: typically –50 to +50. Positive TSB = fresh. Negative TSB = fatigued.
- **What it means**: are you rested (positive) or fatigued (negative)?
- **How to use it**:
- TSB > +10 = very fresh. Good day for hard workout.
- TSB +5 to +10 = fresh. Ready for workouts.
- TSB 0 to +5 = neutral. Can do normal sessions.
- TSB 0 to –10 = slightly fatigued. Consider easy session.
- TSB –10 to –20 = fatigued. Do easy sessions.
- TSB < –20 = very fatigued. Take easy week or rest.
- **Example**: CTL = 70, ATL = 60, TSB = +10 (fresh, ready for hard work). CTL = 70, ATL = 90, TSB = –20 (very fatigued, take easy week).

The Dashboard in Practice:

Monday (Week 1, Early Build Phase):

- TSS yesterday (Sunday long run): 110 TSS.
- CTL: 55 (you're building fitness).
- ATL: 45 (moderate fatigue from weekend).
- TSB: +10 (fresh).
- **Decision**: you're fresh. Tuesday hard bike is good.

Wednesday (Week 1, Mid-Week):

- TSS yesterday (Tuesday hard bike): 100 TSS + (Tuesday evening swim): 50 TSS = 150 TSS total.
- CTL: 55 (unchanged, one day doesn't move it much).
- ATL: 70 (higher; accumulating fatigue from week).
- TSB: –15 (fatigued).
- **Decision**: you're fatigued. Thursday quality session should be easy or skipped. Plan easy runs instead.

Friday (Week 1, End of Week):

- TSS yesterday (Thursday easy run): 30 TSS.
- CTL: 55.
- ATL: 55 (fatigue is dropping as easy sessions accumulate).
- TSB: 0 (neutral).
- **Decision**: you're refreshed a bit. Weekend anchors can be full intensity (long ride/run).

Monday (Week 2, Post-Recovery):

- TSS yesterday (Sunday long run): 110 TSS.
- CTL: 60 (up slightly; a week of training has shifted your fitness).
- ATL: 45 (down; you had easy Friday/Saturday before Sunday).
- TSB: +15 (very fresh).
- **Decision**: great recovery. You're ready to push harder this week.

Using the Dashboard to Prevent Overtraining:

Red Alert (TSB < –20):

- Action: take 3–7 days easy (Level 1). No hard sessions. Just easy Z1.
- Recheck TSB in 3–4 days. Should move toward 0 or positive.

- If still < –20: extend easy period. Something's wrong (sleep, stress, illness, overtraining). Adjust.

Yellow Alert (–20 < TSB < –10):

- Action: reduce volume this week. Drop to Level 2 if you were at Level 3. One quality session max, rest easy.
- Recheck TSB next week. It should move toward 0.

Green (TSB > –10):

- Action: normal training. You're recovered or managing fatigue well.

Key Insight:

- "TSB is your early warning system. If TSB is trending negative, you're heading toward overtraining. Fix it before it breaks you."

SLIDE 52 – When to Adjust (Early Recognition & Response)

SAY:

- Readiness trending down? Reduce volume.
- Cut volume first, not intensity.
- Keep easy movement (don't stop).
- Persistent fatigue? Take 3–7 easy days.

EXPAND:

Indicator 1 – Readiness Trending Down (TSB or Subjective Feel)

- **What it looks like**: TSB has been dropping for 2–3 weeks. Or you're just feeling less "fresh."
- **Why it matters**: downward trend in readiness = cumulative fatigue. You're not recovering between workouts.
- **How to respond**:
- **Week 1 of trend**: monitor closely. Is TSB continuing down, or stabilizing?
- **Week 2 of trend**: start reducing volume (drop to Level 2 if at Level 3). Add an extra easy day.
- **Week 3 of trend**: full easy week (Level 1). Reassess.
- **Action**: "I notice TSB has been –5, –10, –15 over the last three weeks. I'm trending fatigued. This week: easy only. Next week: reassess."

Indicator 2 – Cut Volume First (Don't Cut Intensity)

- **What it looks like**: you're fatigued, but you keep doing hard sessions thinking intensity will "break through" fatigue.
- **Why it's wrong**: hard training adds more stress when your body needs recovery. This leads to overtraining.
- **What to do**: cut volume (drop long sessions from 4h to 2.5h, reduce easy sessions), keep intensity minimal.
- **Why it works**: volume creates cumulative fatigue; intensity is temporary stress. Cutting volume gives recovery room.
- **Action**: "I'm fatigued. Instead of cutting Tuesday hard session, I'll cut Wednesday easy run and Thursday swim. Keep Tuesday, reduce overall volume."

Indicator 3 – Keep Easy Movement (Don't Do Complete Rest)

- **What it looks like**: you're very fatigued and want to stop training entirely.
- **Why complete rest is often wrong**: some movement helps recovery better than zero movement.
- **What to do**: keep doing very easy movement (Z1 only, 30–45 min per session). Walking, easy spin, easy swim.
- **Why it works**: easy movement promotes blood flow, recovery, and prevents complete detraining.
- **Action**: "I'm taking an easy week. But I'll still do 30-min easy runs, 30-min easy swims, 45-min easy spins. Just easy, no intensity."

The Golden Rule – Persistent Fatigue? Take 3–7 Easy Days:

- **Rule**: if you've been fatigued for 2+ weeks, take 3–7 days of easy training (Level 1).
- **What it is**: mostly easy Z1 sessions, zero intensity, zero hard sessions.
- **Duration**: usually 1 week. Sometimes 2 weeks if very fatigued.

- **Outcome**: TSB should move back toward 0 or positive. Readiness should improve.
- **After**: reassess. If fatigued still, extend easy period. If recovered, resume normal training.

Adjust Early, Avoid Setbacks:

- **Best case**: you see fatigue coming (TSB trending down), you adjust early (reduce volume for 1 week), you avoid setback.
- **Worst case**: you ignore fatigue for 3 weeks, you break (illness or injury), you're off training for 2+ weeks. Much worse.
- **The math**: 1 week of preventive easy training saves you from 2+ weeks of forced rest. Worth it.

SLIDE 53 – Common Mistakes (What Derails Most Athletes)

SAY:

- Gray zone: training "moderately hard" too often = chronic fatigue.
- Volume spikes: adding mileage too fast = injury.
- Ignoring body: following plan blindly when body screams "rest" = breakdown.
- Execution errors cost more than fitness errors.

EXPAND:

Mistake 1 – The Gray Zone (Training Moderately Hard Too Often)

- **What it is**: training at intensity that's neither recovery nor hard work. ~80–90% effort, most of the time.
- **Why it's destructive**: gray zone doesn't provide recovery stimulus (easy) or adaptation stimulus (hard). It just fatigues you.
- **Example**: you do 40-min "moderate" runs at 85% effort 4x/week, thinking it's building fitness. Actually, you're never recovering, never getting hard stimulus. Just constant fatigue.
- **The damage**: chronic fatigue, adaptation stalls, overtraining, burnout.
- **How to fix**: 90% of training should be very easy (Z1, recovery). 10% should be hard (Z2–Z3, VO_2). No gray zone.
- **Test**: can you have a full conversation at easy pace? If yes, it's easy. If no, it's too hard. No middle ground.

Mistake 2 – Volume Spikes (Adding Mileage Too Fast)

- **What it is**: jumping from 10 h/week to 15 h/week in one week, or adding 2 hours to long session suddenly.
- **Why it's destructive**: your body needs time to adapt to increased load. Spike it too fast and you get injured.
- **Example**: you do 3-hour long bike for 8 weeks, then suddenly do 5 hours. Your body isn't ready. Injury.
- **The damage**: acute injury, tendonitis, joint pain.
- **How to fix**: increase volume 5–10% per week max. 10 h → 10.5 h → 11 h → 11.5 h (gradual).
- **Rule**: long sessions should increase by 15–30 min every 2–3 weeks, not every week.

Mistake 3 – Ignoring Body (Following Plan Blindly When Your Body Screams "Rest")

- **What it is**: feeling broken (TSB < –20, persistent soreness, dread), but forcing the plan anyway.
- **Why it's destructive**: your body is sending red flags. Ignoring them leads to injury or illness that knocks you out for weeks.
- **Example**: you're exhausted, TSB is –25, but Tuesday hard session is on the plan so you do it anyway. Result: you get sick or injured. Now you're off training for 2 weeks.
- **The damage**: injury, illness, lost training time.
- **How to fix**: listen to your body. If red flags are present, adjust the plan. Flexibility beats rigidity.

Critical Insight – Execution Errors Cost More Than Fitness Errors:

- **Fitness error**: choosing wrong zone structure, suboptimal block order, missing a few sessions. Impact: ~5–10% slower progress.
- **Execution error**: ignoring fatigue, spiking volume, doing gray zone training. Impact: overtraining, injury, forced weeks off. This kills races.
- **The lesson**: getting the system right is 70% of success. Executing it correctly is 30%. But that 30% determines whether you finish strong or DNF (did not finish).

SLIDE 54 – Real Examples (How Integration Works in Real Life)
SAY:
- **Early riser (Sarah)**: 5–6:30am training, family gets rest of day.
- **Splitter (Mike)**: long sessions split (2h + 45m bike instead of 4h).
- **Traveler (Elena)**: run-heavy weeks when traveling, full training when home.

EXPAND:
Example 1 – The Early Riser (Sarah, Lawyer)
- **Life constraint**: full-time law job, unpredictable hours, spouse and two kids.
- **Solution**: "Done by 6 am. Evenings free."
- **Strategy**: wake up 5 am, train 5–6:30 am (bike or run), home by 6:30 am before kids wake, coffee with family, off to work.
- **Weekly structure**:
- Mon: off (sleep in).
- Tue: 5–5:45 am hard bike, then commute work.
- Wed: 5–5:45 am swim (pool near office).
- Thu: 5–5:45 am quality run.
- Fri: off (sleep in, family morning).
- Sat: 6–9 am long bike (trainer or outside, kids have dad-time with partner).
- Sun: 7–8:30 am long run (partner handles kids).
- **Total**: ~12 hours/week.
- **Why it works**: early mornings are non-negotiable. Training is done before life starts. Evenings are family time. Partner fully supports because they see it.
- **The key**: consistency of wake-up time. Same 5 am every day (except Wed/Fri rest) = automatic.

Example 2 – The Splitter (Mike, Dad of Three)
- **Life constraint**: young kids, wife works, chaos at home, limited windows.
- **Solution**: "Split long rides. Fits around kids."
- **Strategy**: can't do 4-hour unbroken ride. So: 2-hour bike ride Saturday morning (wife has kids). Then 45-min easy bike Sunday morning (kids tag along). Long run Sunday afternoon (kids watch or play nearby).
- **Weekly structure**:
- Mon–Fri: early morning runs (30–45 min, 4:30–5:15 am before kids wake). Commute bike counts as training (20–30 min).
- Sat: 2-hour bike (alone, early morning, wife handles kids).
- Sun: 45-min easy bike (kids come along), then 90-min long run afternoon (wife handles kids).
- **Total**: ~8 hours/week.
- **Why it works**: training is split into multiple short sessions instead of one long block. Early mornings protect training. Family is involved when possible.
- **The key**: acceptance that long sessions are split. No pride about solo 4-hour rides. Split rides still build fitness.

Example 3 – The Traveler (Elena, Business Owner)
- **Life constraint**: unpredictable schedule, travel monthly, meetings that move, no consistency.
- **Solution**: "Run-heavy block. Minimum dose."
- **Strategy**: bikes and pools are travel-dependent. Running is always available. So: emphasize running when traveling, skip hard bike work, add flexible blocks.
- **Weekly structure**:
- **Normal weeks** (home): long bike Saturday, long run Sunday, 2–3 quality sessions, 2–3 easy swims = 12 h/week.
- **Travel weeks** (away): easy runs Monday–Friday (30–60 min each), no bike/swim, long run Saturday or Sunday = 4–6 h/week. Acceptance that travel week is reduced.
- **Post-travel weeks** (back home): ramp back up to 12 h/week.

- **Pattern over 4 weeks**: normal 12h, travel 5h, normal 12h, normal 12h = average 12.75 h/week. Still building fitness despite travel.
- **Why it works**: recognizes travel disrupts routine. Has a plan (run-heavy) for travel weeks. Doesn't try to maintain full volume while traveling.
- **The key**: flexibility blocks. Adjust training around life, not against it.

Common Themes Across All Three:

- All three protect anchor sessions (long bike, long run) as much as possible.
- All three have realistic volume (8–12 h/week), not fantasy 20+ h.
- All three adapted the system to fit their life, not forced their life to fit the system.
- All three succeeded (finished strong, didn't burn out, sustainable for years).

SLIDE 55 – Understanding the Timeline (Aerobic Gains Take Weeks, Durability Takes Months)

SAY:

- Aerobic gains take weeks.
- Durability takes months.
- Short disruptions rarely erase long-term progress.
- Steady stimulus over months beats perfect single weeks.

EXPAND:

Week 1–3: Foundation Building

- **What's happening**: your body is learning consistency. Aerobic system is starting to adapt. Durability is building but not apparent yet.
- **What you notice**: first few sessions feel hard. You're sore. Recovery is slow.
- **Progress markers**: fitness isn't obvious yet. Don't expect it.
- **Mental game**: this is the "faith phase." You have to trust the process.
- **Duration**: lasts 3 weeks minimum.

Week 4–8: Adaptation Phase

- **What's happening**: aerobic adaptations are occurring (mitochondrial growth, capillary expansion, fat-burning efficiency improving). Fitness is starting to show.
- **What you notice**: sessions that were hard at week 1 now feel moderate. Recovery is improving. Power/pace is increasing slightly.
- **Progress markers**: VO_2 is improving. Heart rate is lower for the same effort. Durability is noticeably better.
- **Mental game**: this is motivating. You're seeing progress. Keep going.
- **Duration**: lasts 5 weeks (weeks 4–8 of training).

Week 8–12: Building Phase

- **What's happening**: bigger fitness gains. Durability is solidifying. You can handle higher training loads without breaking.
- **What you notice**: longer sessions feel sustainable. Recovery is fast. Performance is clearly better.
- **Progress markers**: power is up 5–10%. Pace is faster. TSB management is easier (you're not getting deeply fatigued).
- **Mental game**: confidence is building. You're believing in the plan now.
- **Duration**: lasts 5 weeks (weeks 8–12 of training).

Week 12–16: Durability Solidifying

- **What's happening**: durability is now solid. You can repeat hard weeks with consistent recovery. Body handles training stress well.
- **What you notice**: long sessions no longer feel risky. You're mentally strong. Performance is holding or improving.
- **Progress markers**: durability markers (long sessions feel steady, recovery is fast) are green. Confidence is high.
- **Mental game**: you're in the zone. Training feels good. Racing feels possible.
- **Duration**: lasts 5 weeks (weeks 12–16 of training).

Week 16–20: Peak Building

- **What's happening**: durability is very solid. You're now adding intensity. Peak power/speed work happens here. VO2 training is effective.
- **What you notice**: race-pace work feels possible. You're fast. Peak is visible.
- **Progress markers**: race pace feels controlled. Pacing and fueling are dialed. Confidence is very high.
- **Mental game**: you're ready. The race feels achievable.
- **Duration**: lasts 5 weeks (weeks 16–20 of training).

Week 20–24: Race Prep & Taper

- **What's happening**: you're executing race simulations. Durability is proven. You're now just maintaining fitness while recovering.
- **What you notice**: long sessions feel easy-ish. You're resting more. Race is feeling near.
- **Progress markers**: no more fitness building needed. You have what you need. Just stay fresh.
- **Mental game**: final confidence surge. You're ready.
- **Duration**: lasts 5 weeks (weeks 20–24 of training).

Key Insight – Why Patience Matters:

- **Week 4**: "I've trained for 4 weeks and I'm not any faster. Does this work?" (You're still in foundation building.)
- **Week 8**: "I've trained for 8 weeks and I'm noticeably fitter. The system works." (Foundation is building, adaptations are happening.)
- **Week 12**: "I've trained for 12 weeks and I'm strong and durable. I'm confident." (Durability is solidified.)
- **Week 16–20**: "I've trained for 16–20 weeks and I'm peak fit. I'm ready to race." (Peak is reached.)

The Mantra:

- "Aerobic gains take weeks. Durability takes months. Short disruptions rarely erase long-term progress. Steady stimulus over months beats perfect single weeks."

SLIDE 56 – The Stability Index (Why It Matters)

SAY:

- Stability = week-to-week variance in training load.
- Ideal: < 10% variance.
- Low stability (boom/bust): injury risk, progress stalls.
- High stability (steady): consistent progress, sustainable.

EXPAND:

What is Stability Index?

- **Definition**: week-to-week variance in your training load. How much does volume change from week to week?
- **Ideal**: <10% variance. Week 1 = 10 hours. Week 2 = 9–11 hours (not 5 or 15).
- **Why**: your body adapts to consistent stimulus. If stimulus varies wildly, your body is always re-adapting and never fully adapts.

Low Stability (Boom/Bust Cycle):

- **What it looks like**: Week 1 = 8 hours. Week 2 = 15 hours (boom). Week 3 = 4 hours (bust). Week 4 = 12 hours (boom again).
- **Why it happens**: life is unpredictable. One week has time, next week doesn't. No structure.
- **Result on body**:
- Week 1 (8h): body is adapting to moderate load.
- Week 2 (15h): sudden spike. Body is shocked. Fatigue accumulates fast.
- Week 3 (4h): sudden drop. Body isn't recovering; it's detraining.
- Week 4 (12h): boom again. Body is re-adapting.
- **Overall**: body is never in steady adaptation. It's always swinging between shock and detraining.
- **Negative outcomes**: fatigue spikes, injury risk rises, progress stalls or regresses, burnout.
- **Example athlete**: weekend warrior who does nothing Mon–Thu, then massive workouts Fri–Sun. Boom/bust every week.

High Stability (Steady, Repeatable Cycle):

- **What it looks like**: Week 1 = 10 hours. Week 2 = 10.5 hours. Week 3 = 9.5 hours (tiny variance). Week 4 = 11 hours. Weeks average 10.2 h/week.
- **Why it happens**: planned consistency. You're designing training to be repeatable.
- **Result on body**:
- Week 1–4: body experiences consistent stimulus (9.5–11 h/week). Adaptations are steady.
- Every week, body knows what's coming: same structure, same load, minor variations only.
- Adaptations compound: week 2 benefits from week 1, week 3 from weeks 1–2, etc.
- **Overall**: body is in steady adaptation. It's improving linearly.
- **Positive outcomes**: consistent progress, no surprise fatigue, lower injury risk, sustainable for months.
- **Example athlete**: Sarah (lawyer) who does 12 h/week every single week. Stability is 95%. Progress is steady.

Why Stability Matters for Long-Course Racing:

- Long-course requires durability. Durability is built through steady, consistent stimulus over months.
- High stability = body adapts predictably = durability builds = you can handle race day.
- Low stability = body is always recovering/re-adapting = durability doesn't build = race day feels overwhelming.
- Racing is stable stress (long, steady effort). Training should mirror that (stable, steady load week-to-week).

How to Build Stability:

1. **Design your weekly structure** (anchor sessions locked in, easy sessions filling gaps).
2. **Keep week-to-week variance <10%** (if 10h one week, aim for 9–11h next week, not 5 or 15).
3. **Use Plan A/Plan B** to flex within that band (Plan A is 11h, Plan B is 8h, when life happens use Plan B, but keep variance small).
4. **Repeat the same weekly structure** (Monday rest, Tuesday hard bike, Sat long bike, etc.) becomes automatic.

How to Monitor Stability:

- Plug your weekly training hours into a spreadsheet for the last 8 weeks.
- Calculate variance (how much do hours swing from week to week?).
- Ideal: <10% variance.
- If variance is 20%+, you're in boom/bust cycle. Design for consistency.

SLIDE 57 – High Stability vs Low Stability (Visual Comparison)

SAY:

- Left: low stability, spiky chart (boom/bust).
- Right: high stability, steady chart (steady line trending up).
- Both athletes do ~100 hours over 10 weeks.
- Low-stability athlete's body is constantly shocked; high-stability athlete's body is steadily adapting.

EXPAND:

Low Stability Chart (Boom/Bust):

- Week 1: 10 hours.
- Week 2: 5 hours (drop, reason: busy week).
- Week 3: 15 hours (boom, reason: trying to catch up).
- Week 4: 6 hours (bust, reason: exhausted from week 3).
- Week 5: 14 hours (boom again).
- Week 6: 7 hours (bust again).
- Week 7–10: continues spiky pattern.
- **Total over 10 weeks**: ~100 hours.
- **Pattern**: wild swings, no consistency.

- **Body response**: constantly adapting to new stimulus, never fully adapting, high fatigue, high injury risk, low progression.

High Stability Chart (Steady & Repeatable):
- Week 1: 10 hours.
- Week 2: 10.5 hours (consistent, slightly higher).
- Week 3: 9.5 hours (dip, planned recovery, still in band).
- Week 4: 11 hours (back up, still in band).
- Week 5: 10 hours (back to normal).
- Week 6–10: continues steady pattern, averaging 10.2 h/week.
- **Total over 10 weeks**: ~102 hours (nearly same as low-stability).
- **Pattern**: steady line, minor ups and downs, all within 9–11 h band.
- **Body response**: consistent stimulus, steady adaptation, lower fatigue, lower injury risk, high progression.

Visual Insight:
- Both athletes train ~100 hours over 10 weeks (same total).
- Low-stability athlete ends up exhausted, injured, fitness is plateaued or down.
- High-stability athlete ends up strong, healthy, fitness is up.
- The difference isn't the total hours; it's how those hours are distributed.
- Steady distribution = better adaptation.

Real-World Example:
- **Low Stability Athlete**: "I'll crush it this week (15 h), then recover next week (5 h)." After 10 weeks, they're burned out.
- **High Stability Athlete**: "I'll do 10 h/week every week, varying slightly if life demands." After 10 weeks, they're stronger.

The Mantra:
- "Steady stimulus over months beats spiky stimulus. Stability builds durability."

SLIDE 58 – Minimum Effective Training (Quality > Junk Volume)

SAY:
- **70.3**: 8–12 h/wk (9–11h typical), 90% easy, 10% hard.
- **Ironman**: 10–15 h/wk (11–14h typical), 90% easy, 10% hard.
- Quality > junk. Consistency > heroics.

EXPAND:

70.3 Minimum Effective Training:
- **Weekly hours**: 8–12 hours per week.
- **Typical structure**:
- Saturday long bike: 2.5–3.5 hours (Z1–Z2).
- Sunday long run: 75–90 min (Z1–Z2).
- One quality session: bike or run, 45–60 min (Z2–Z3).
- 2–3 easy sessions: 30–60 min each (Z1), spread through week.
- **Total**: ~9–11 hours.
- **Distribution**: 85–90% easy (Z1–Z2), 10–15% hard (Z2–Z3).
- **Why it works**:
- Long bike/run build durability (foundation and walls).
- One quality session maintains sharpness and threshold.
- Easy sessions fill volume without fatigue.
- Race is 70% race pace (Z2) and 30% faster. This training mirrors that.

Ironman Minimum Effective Training:
- **Weekly hours**: 10–15 hours per week.
- **Typical structure**:
- Saturday long bike: 3.5–4.5 hours (Z1–Z2).
- Sunday long run: 90–120 min (Z1–Z2).
- One quality session: bike or run, 45–60 min (Z2–Z3).

- 3–4 easy sessions: 30–60 min each (Z1), spread through week.
- **Total**: ~11–14 hours.
- **Distribution**: 85–90% easy, 10–15% hard.
- **Why it works**:
- Long sessions build massive durability (race is 10–17 hours).
- Easy sessions fill volume sustainably.
- One quality session maintains threshold; prevents complete aerobic-only training.
- Ironman is 95% race pace (easy Z1–Z2) and 5% faster. This training is 90% easy, which is close.

Quality > Junk Volume:

- **Quality**: sessions that have a purpose and create adaptation. Z1 (recovery), Z2 (threshold), Z3 (lactate threshold), VO_2 (power/speed).
- **Junk volume**: sessions done without purpose. Gray zone efforts (75–85% intensity, neither easy nor hard), long easy rides that don't teach anything, random efforts.
- **Example**:
- Quality week: 10 hours = 2h easy (Z1), 3h long bike (Z1–Z2), 90m long run (Z1–Z2), 45m threshold bike (Z3), 30m easy swim (Z1), extras.
- Junk week: 10 hours = 60m moderate bike (gray zone), 75m moderate run (gray zone), 60m moderate bike (gray zone), 60m moderate run (gray zone), plus random stuff. All gray zone, no stimulus, no recovery.
- Quality week: adaptation happens. Junk week: fatigue accumulates without progress.

Consistency > Heroics:

- **Consistency**: same ~10 h/week every week for 12 weeks = 120 hours total = huge fitness gain.
- **Heroics**: 15 h/week for 4 weeks (60 hours) + missing 8 weeks due to burnout = 60 hours total. Less fitness despite higher individual weeks.
- **The math**: 52 consistent weeks (one per year) beat 4 perfect weeks + 48 off weeks.

How to Identify Minimum Effective Training for Your Goal:

- **Goal: Finish 70.3**: 8–10 h/week is minimum. Gets you fit enough to finish strong.
- **Goal: PR 70.3**: 10–12 h/week. Higher volume, more intensity.
- **Goal: Finish Ironman**: 10–12 h/week is minimum. Gets you fit enough.
- **Goal: PR Ironman or qualify for Worlds**: 12–15 h/week. Higher volume, strategic intensity.
- **Key insight**: the higher your goal, the more volume you need. But even high goals don't need 20+ h/week; 12–15 h/week is usually sufficient.

Practical Application:

- Don't do 15 h/week if 12 h/week is sufficient for your goal. The extra 3 hours add injury risk and burnout risk without significant benefit.
- Do the minimum that achieves your goal, then stop. Efficiency.
- If you're at 12 h/week and feeling strong, don't add more just to "get stronger." You're already strong enough.

SLIDE 59 – Consistency Looks Different (Recap)
SAY:

- Typical Plan (rare): perfect linear progress (almost never happens).
- Life Happens (common): ups, downs, misses (realistic, still works).
- Weekend Warrior (risky): massive spikes, injury prone.
- Steady (ideal): repeatable load week-to-week (sustainable).
- All four athletes can finish. Adapt to your reality.

EXPAND:
Pattern 1 – Typical Plan (RARE – Perfect Linear Progress)

- Week-to-week progression is linear. Every week, volume increases smoothly.
- Example: Week 1 = 8h, Week 2 = 9h, Week 3 = 10h, etc.
- Reality: almost never happens. Life gets in the way.
- **Outcome**: perfect fitness progression if you can pull it off. But most can't.

Pattern 2 – "Life Happens" (COMMON – Ups, Downs, Misses)

- Some weeks go to plan, some weeks don't. Ups and downs. But overall forward trend.
- Example: Week 1–3 solid, Week 4 work crisis (cut to Plan B), Week 5–6 back to normal, Week 7 family issue (miss a day or two), Week 8–12 strong.
- Reality: most athletes follow this pattern.
- **Outcome**: slower than perfect plan, but still progress. Sustainable because it adapts.

Pattern 3 – "Weekend Warrior" (RISKY – Boom/Bust Spikes)

- Mon–Thu almost nothing. Fri–Sun massive workouts.
- Example: off Mon–Thu, then Friday = big bike, Saturday = long ride, Sunday = long run.
- Reality: spiky load, inconsistent, high injury risk.
- **Outcome**: burnout or injury before race day. High risk.

Pattern 4 – "Steady" (IDEAL – Moderate, Frequent, Adaptable)

- Similar volume every week (±10% variance). Consistent structure. Slight adjustments as needed.
- Example: Week 1–12 all = 10–12 h/week, same structure, minor tweaks based on life.
- Reality: doable if you design it and protect it.
- **Outcome**: steady progress, sustainable, high success rate.

Key Insight:

- "Consistency doesn't look the same for everyone. Adapt the pattern to your reality."
- Sarah = Steady pattern (lawyer, predictable schedule).
- Mike = Life Happens pattern (chaotic with young kids, but usually recovers).
- Elena = Steady-to-Life Happens pattern (switches between depending on business demands).
- All succeed because they adapt.

SLIDE 60 – Fun Improves Consistency (Enjoyment Drives Habit)
SAY:

- Enjoyment → consistency → durability → confidence (loop).
- Create rituals you look forward to.
- Train socially when possible.
- Celebrate small wins daily.
- Fun is the x-factor.

(Content same as provided earlier)

SLIDE 61 – What Success Looks Like (Consistent, Not Perfect)
SAY:

- Consistent, not perfect.
- 85–90% adherence over 12–24 weeks.
- Protect high-value sessions (anchors).
- Adapt quickly when life happens.
- Maintain stability (low variance).

(Content same as provided earlier)

SLIDE 62 – The Confidence Gap (You're Motivated, But Also Unsure)
SAY:

- Questions: Am I doing enough? Will I finish? What if life derails me?
- Gap exists because: fear, uncertainty, comparison, imposter syndrome.
- Gap closes through **familiarity**, not more training.

(Content same as provided earlier)

SLIDE 63 – Confidence Comes From (Pacing, Fueling, Familiarity)
SAY:

- **Pacing**: practiced race pace 10+ times. You know you can do it.
- **Fueling**: practiced race nutrition on every long session. Stomach is ready.

- **Familiarity**: practiced race conditions 10+ times. Nothing surprises you.

(Content same as provided earlier)

SLIDE 64 – Confidence Loop (How Familiarity Builds Belief)
SAY:

- Consistency → base work → durability → execution → race sim → **confidence**.
- Each step feeds the next. Six steps from nervous to ready.

(Content same as provided earlier)

SLIDE 65 – Where You Are (Right Now, Same Principles, Different Timelines)
SAY:

- 70.3 PA athletes (~9 weeks out): Peak consistency phase. Stack solid weeks. No heroics. Protect anchors. Practice race pace and fueling on every long session. Goal: consistency.
- IM Lake Placid athletes (~13 weeks out): Durability-building phase or early build phase. Focus on long sessions. Introduce one quality session. Practice race pace with fueling. Goal: durability + execution prep.
- **Both groups**: non-negotiables don't change: consistency, protect anchors, test zones, practice nutrition, maintain stability.
- **Time is tight**: both groups have limited time. Make every week count. No wasted sessions. Quality over junk.

(Content same as provided earlier)

SLIDE 66 – Two Races, Same Rules (The Non-Negotiables)
SAY:

- Non-negotiable 1: Protect anchor sessions.
- Non-negotiable 2: 90% training in ZR–Z2 (easy).
- Non-negotiable 3: Execute minimum-effective week (Level 1, 2, or 3).
- Non-negotiable 4: Plan B when life happens.

(Content same as provided earlier)

SLIDE 67 – 70.3 PA – 9 Weeks Out (Peak Consistency Phase)
SAY:

- Phase: peak consistency. Stack solid weeks. Practice race execution.
- Anchors: 2.5–3.5h bike, 75–90m run, every week.
- Goal: 9–12 h/week consistency.
- If life happens: Level 1 (keep anchors, cut filler).

(Content same as provided earlier)

SLIDE 68 – Lake Placid IM – 13 Weeks Out (Durability Building Phase)
SAY:

- Phase: durability building. Patient, progressive loading.
- Anchors: 3.5–4.5h bike, 90–120m run, every week.
- Goal: 11–15 h/week, progressing.
- Finish tired, not empty.
- If life happens: Level 1 (keep anchors, cut filler).

(Content same as provided earlier)

SLIDE 69 – The Common Thread (Success Across Different Timelines)
SAY:

- Thread 1: Hit anchor sessions (regardless).
- Thread 2: Respect intensity (90% easy, 10% hard).
- Thread 3: Minimum-effective week (choose level).
- Thread 4: Plan B (flexibility).

- Success metric: consistent execution, not perfection.

(Content same as provided earlier)

SLIDE 70 – Race-Day Execution (Pacing & Nutrition Strategy)
SAY:
70.3 PA:

- **Bike**: 78–82% CP. Run: build effort Z2–Z3. Fuel: 60–80g carbs/h.

IM Lake Placid:

- **Bike**: 65–70% CP. Run: Z2 jog. Fuel: 80–100g carbs/h.

Both:

- Golden rules: no new gear, test breakfast, train gut, respect taper.

(Content same as provided earlier)

SLIDE 71 – Sample Week in Action (Real Structure & Flow)
SAY:

- Mon: rest. Tue: hard bike. Wed: easy run + swim. Thu: easy bike + strength. Fri: swim + easy run.
- Sat: long bike (anchor). Sun: long run (anchor).
- Total: 10–11h. Repeatable. Rhythm.

(Content same as provided earlier)

SLIDE 72 – Final Weeks Protocol (Volume Reduction Strategy – Taper)
SAY:

- 2 weeks out: cut to 75% volume.
- Race week: cut to 50% volume.
- Keep easy movement (don't stop entirely).
- Golden rule: taper intensity, not just movement.

EXPAND:
The Taper Concept:

- **Traditional taper**: reduce volume massively (50%+), stop training entirely for the final days.
- **Result**: athletes arrive race day rested but detrained. They've lost sharpness.
- **Better taper**: reduce volume moderately (75% volume 2 weeks out, 50% volume race week), keep some intensity and frequency (but shorter, lower-intensity).
- **Result**: athletes arrive race day fresh and sharp. They maintain fitness while recovering.

The 2-Week Drop (2 Weeks Out from Race):

- **Normal week**: 12 hours training.
- **Taper week 1** (2 weeks out): cut to 75% = ~9 hours.
- Saturday long bike: 1.5–2 hours (shorter than normal).
- Sunday long run: 60 minutes (shorter than normal).
- One quality session (Tuesday or Thursday): 30 min threshold work (shorter, not max effort, just maintaining).
- Easy sessions: 2–3x, 30 min each (same frequency, shorter duration).
- Total: ~9 hours, mostly easy, one short quality session.
- **Why it works**: volume is lower (reducing fatigue), frequency is same (maintaining habit), one quality session maintains sharpness.

The Race Week Taper (Race Week):

- **Race week**: cut to 50% = ~6 hours.
- Mon/Tue: easy spin or jog (15–20 min each, just movement).
- Wed: light swim or nothing.
- Thu: easy spin or run (20 min, shakeout).
- Fri: very easy jog or walk (15 min), logistics practice (test transitions, check gear).
- Saturday morning (race day for some): nothing or 15 min very easy jog as warmup.
- Sunday (race day for others): nothing, just rest and prepare.

- **Total**: ~4–6 hours, all easy.
- **Why it works**: volume is very low (maximum recovery), but you're still moving (no detraining), logistics are practiced.

The Golden Rule – "Resting Too Much Kills Momentum. Taper Intensity, Not Just Movement."

- **Wrong approach**: "I'll do zero training the final 3 days." Result: you arrive race day stale, no warm blood, movements feel clunky.
- **Right approach**: keep moving (short, easy sessions), just reduce intensity and duration. You arrive race day fresh, warm, loose.
- **Example**:
- **3 days before race**: 20 min easy jog. Just moving, feeling good.
- **2 days before race**: 30 min easy bike. Spinning, keeping the legs happy.
- **1 day before race**: nothing or 15 min easy walk. Just recovery.
- **Race morning**: 15–20 min very easy jog as warmup. Blood is flowing, body is warm.

What You Don't Do in Final Weeks:

- No new workouts.
- No hard sessions.
- No testing.
- No trying new gear or nutrition.
- No massive training volumes.

What You Do Do in Final Weeks:

- Keep easy movement.
- Maintain some frequency (move most days, just not hard).
- Practice transitions and logistics.
- Practice race-day breakfast and gear.
- Rest well (prioritize sleep).
- Manage anxiety (movement helps, planning helps).

Mental Challenge – The Taper (Why It's Hard):

- Most athletes feel antsy during taper. They want to train hard to "get in one last hard session."
- But hard training now = fatigue on race day. It's the opposite of what you want.
- **Mindset**: "I've trained for 12 weeks. I've built fitness. The work is done. Now I rest and recover. Rest is active (easy movement). Trust the training."
- **Goal**: arrive race day fresh, confident, ready to execute.

SLIDE 73 – Execution vs Fitness (What You HAVE vs What You DO)

SAY:

- **Fitness**: what you have (engine, zones, capacity).
- **Execution**: what you do (pacing, fueling, decisions).
- Fitness is potential. Execution is reality.
- In final week, can't build more fitness, but can improve execution.

EXPAND:

FITNESS – "What You HAVE"

- **Definition**: the aerobic engine you've built through training.
- **Measures**:
- Engine potential (VO_2 max, max power, anaerobic power).
- Physiology zones (CP, CS, CSS).
- Raw capacity (how much stress your body can handle).
- **Built**: over months of consistent training. Base work, threshold work, peak work.
- **Examples**: your CP is 280W. Your critical speed is 7:00/mile. Your VO_2 max is high.
- **Fixed**: you can't change fitness on race day. It's what you have.

EXECUTION – "What You DO"

- **Definition**: the decisions you make and actions you take during the race.
- **Measures**:
- Pacing strategy (how hard you bike, how hard you run, when you push/hold back).

- Nutrition timing (when you eat, how much, what you eat, managing fuel intake).
- Mental resilience (staying calm when things hurt, pushing through hard moments, pacing discipline).
- Transitions (clean bike-to-run, gear management, logistics).
- **Made**: during the race in real time. Every moment offers a decision.
- **Examples**: you're 10 min into the run and your legs are heavy. Do you panic and slow down? Or stay disciplined and hold pace? That's execution.
- **Changeable**: you can improve execution in the final week (mindset, practiced pacing, logistical prep).

Why Execution Matters:

- **Scenario 1 (High Fitness, Poor Execution)**:
- You've trained well. Your fitness is high (CP = 300W, CS = 6:50/mile).
- But on race day, you panic and go out too hard on the bike. By km 50, you're cooked.
- You hit the run unable to run hard. You jog slow and finish a 5:30 70.3 when you could have done 4:45.
- **Result**: fitness wasted due to poor pacing execution.
- **Scenario 2 (Moderate Fitness, Excellent Execution)**:
- You've trained decently. Your fitness is moderate (CP = 260W, CS = 7:15/mile).
- On race day, you pace perfectly. You hold 78% CP on the bike for the full 90 min. You're controlled on the run.
- You race smart, fuel perfectly, manage the mental challenge.
- You finish a 5:15 70.3, nearly achieving a time that seemed impossible for your fitness level.
- **Result**: fitness amplified by excellent execution.

Fitness is Potential. Execution is Reality.

- You can be fit but race poorly (fitness not realized).
- You can be moderately fit but race well (fitness maximized through execution).
- The gap between potential and realized performance = execution.

Why You Should Care:

- You've trained 12 weeks. You've built fitness.
- Now, in the final week, you can't build more fitness. That's done.
- But you can improve execution: finalize pacing strategy, test gear, practice mental resilience, solidify fueling.
- Execution improvements directly improve race outcome.

How to Improve Execution (This Week):

1. **Finalize pacing strategy**: Know your numbers (bike pace, run pace, limits). Write it down.
2. **Practice transitions**: Do a full dry-run on a training day (bike ride ending in a 20 min run with transition).
3. **Test race nutrition one more time**: Do a final long session with exact race fueling.
4. **Build mental toughness**: Visualize race day. Imagine hard moments. Practice staying calm.
5. **Logistics**: know race-day schedule, know transitions, know the course (if possible).

SLIDE 74 – Module 05 Takeaways (Consistency and Patience)

SAY:

- Consistency beats perfection.
- Durability takes months to build.
- Metrics help you track progress.
- Confidence is trained through six steps.
- By race day, you'll be ready.

EXPAND:

- Consistency is the foundation of long-course training. 12 good weeks beat 4 perfect weeks.
- Progress markers (green and red flags) tell you if you're on track or need to adjust.
- Metrics (TSS, CTL, ATL, TSB) give you data to make smart decisions.
- Timelines are real: aerobic gains take weeks, durability takes months.

- Stability (low variance week-to-week) allows steady adaptation.
- Fun drives consistency. Build rituals and train socially when possible.
- Success looks like 85–90% adherence, not perfection.
- Confidence is trained through repetition: practice pacing, practice fueling, practice transitions.
- The confidence loop is: consistency → base → durability → execution → race sim → confidence.

SLIDE 75 – Key Takeaways (Remember These Rules)
SAY:
1. **Consistency beats perfection** (12 good weeks > 4 perfect weeks).
2. **Test, don't guess** (zones are your GPS).
3. **Protect the anchors** (long ride + long run are sacred).
4. **Train your gut** (practice race nutrition).

(Content same as provided earlier)

SLIDE 76 – More Key Takeaways (Summary & Critical Success Factors)
SAY:
- The four non-negotiables.
- Your success formula: STRUCTURE + WORK + FLEXIBILITY = RESULT.

(Content same as provided earlier)

SLIDE 77 – The Full-Life Adjustment Rule (When Life Stress Rises)
SAY:
1. Reduce volume (not intensity).
2. Keep light movement.
3. Protect key sessions.
4. Resume gradually.

(Content same as provided earlier)

SLIDE 78 – Your Action Plan (7-Day Starter Blueprint)
SAY:
- **Day 1–2**: Your race is already set (70.3 PA ~9 wks, IM Lake Placid ~13 wks). Lock it in mentally.
- **Day 3**: Family talk. "Let's plan together."
- **Day 4–5**: Map anchors. Lock in calendar 8–12 weeks out.
- **Day 6–7**: Test zones THIS WEEK. Bike CP (Sat). Run CS (Fri or Sat).
- **Week 2**: Pick your level (most: Level 2 = 9–11h). Execute one full week.
- **Week 3–4**: Repeat. Add fueling practice (Saturday long bike: practice gels/bars). Family check-in. Track metrics.
- **By day 30**: Rhythm locked, zones known, family on board, ready to train hard.

EXPAND:
Day 1–2 – The Big Decisions
- **Choose your race**: Your race is already set. 70.3 PA in ~9 weeks, or IM Lake Placid in ~13 weeks. Lock it in mentally.
- **Understand your timeline**: you have 8–12 weeks to race day. That's your runway. It's tight but doable if you start now.
- **Commit**: write it down, tell someone, make it real. This is your north star.
- **Why**: clarity on your goal clarifies everything else. Your training plan, your family talk, your weekly structure—all flow from this decision.

Day 3 – The Family Talk
- **Schedule 30 minutes** with your partner/family (tonight or tomorrow).
- **Use the script**: "I want to train for [race name] on [date]. I know it's coming up soon (8–12 weeks). It's important to me. But not at our expense. I want us to figure this out together. What questions do you have? What concerns you?"

- **Listen**: hear their concerns (will you be crazy-busy? are you putting your health at risk? will I see you?).
- **Collaborate**: set “untouchable” family time zones. Show how training integrates with life.
- **Be realistic**: “I’m training 10–12 hours/week, but most of it is early morning or weekends. Weekday evenings are ours. Saturday after my ride is family time.”
- **Outcome**: family buys in. They understand it’s short-term (8–12 weeks, then you’re done for a bit). They become supporters instead of resentful.

Day 4–5 – Map Weekly Anchors

Identify your anchor sessions:

- **Long bike day**: pick one day (usually Saturday). What time works? Example: Saturday 6–9 am (home by 9:30 am).
- **Long run day**: pick one day (usually Sunday, 24h after long bike). Example: Sunday 7–8:30 am.
- **Quality session day**: pick one day (usually Tuesday or Thursday evening). Example: Tuesday 5:30–6:15 pm.
- **Easy sessions**: identify 2–3 gaps in your week (early mornings, lunch breaks, evening). Example: Mon 6–6:45 am run, Wed 12–12:30 pm swim, Fri 5:30–6 pm easy jog.

Lock these into your calendar:

- **Calendar all 8–12 weeks now**. Don’t wait. Lock in every Saturday long bike, every Sunday long run.
- **Tell your family**: “These are my training times. I’ve blocked them. Outside these times, I’m yours.”
- **Why**: locked calendar = protected training. It becomes non-negotiable, like a work meeting. Less likely to get disrupted.

Day 6–7 – Schedule First Tests

Test your zones immediately (this week or next):

- **Bike CP test**: 3-min all-out + 12-min all-out (same session with recovery between, or split over two days). Saturday or Sunday morning before long sessions. Flat road or trainer.
- **Run CS test**: 400m + 800m + 3200m all-outs. Track or GPS flat route. Could be Friday evening or Saturday morning (different day from bike test).
- **Swim CSS test** (optional): 400m all-out + 1500m all-out (48h apart). Find a pool. Could be next week.
- **Why now**: you need zones locked in before Week 1 of training. Zones drive every workout. Without them, you’re guessing.
- **Timeline**: test this week (Days 6–7) or next week (Days 8–14). Get results by Week 1 so you can use them.

Action items:

- Book a flat road or trainer for bike test (Saturday 7 am, 60 min total, with warm-up/cool-down).
- Book a track or GPS route for run test (Friday evening or Saturday, 30 min total).
- Find a pool for swim test (optional; could do Week 2 if not urgent).
- **Input results into training app** (TrainingPeaks, Strava, etc.) immediately after testing so zones auto-calculate.

SLIDE 79 – Questions? Let’s Connect (Next Steps & Contact)

SAY:

- Website: angelanaethcoaching.com
- Email: info@angelanaethcoaching.com
- Free consult: 20 min call to discuss your race.

CLOSE WITH:

- **Consistency beats perfection.**
- **Integration beats isolation.**
- **Adaptation beats rigidity.**
- **Action beats anxiety.**

You have 8–12 weeks. That’s enough. Start this week. I’ll see you at the finish line.

EXPAND:

How to Reach Us:

- **Website**: angelanaethcoaching.com (resources, coaching info, free articles, zone calculators).
- **Email**: info@angelanaethcoaching.com (questions, coaching inquiries, race stories).
- **Schedule a consultation**: 20-minute free call to discuss your race, your timeline, and your training approach.
- **Be specific in your email**: mention your race date, your current weekly hours, what support you need.

What We Offer:

- **Personalized coaching**: one-on-one custom plans tailored to your 8–12 week timeline, weekly check-ins, data analysis, real-time adjustments.
- **Group programs**: cohort-based training groups (8–12 week sprints for 70.3/IM), community support, group Q&A sessions.
- **Free resources**: downloadable training templates (Level 1/2/3 week templates), zone calculators, nutrition guides, 7-day starter blueprints (all on website).
- **Quick-start packages**: for athletes who are 8–12 weeks out and need immediate structure (pre-built plans, no coaching).
- **Community**: access to athlete forum, race-day Q&A, post-race debrief, celebration.

The Process (How We Work Together):

1. **Reach out**: email info@angelanaethcoaching.com or schedule a call at angelanaethcoaching.com/consult.
2. **Quick assessment**: 20-min call to understand your race, timeline, life, goals.
3. **Build your plan**: customized 8–12 week progression based on where you are now.
4. **Execute together**: weekly check-ins (if full coaching) or self-guided with resources (if quick-start package).
5. **Race day**: you execute the plan, we support, you finish strong.
6. **Celebrate**: you cross that finish line. We celebrate with you.

Special Note for 8–12 Week Athletes:

- You're in crunch time. That's okay. Eight weeks is enough if you start now.
- Get your zones tested this week.
- Lock in your calendar this week.
- Family talk happens this week.
- By next week, you're training.
- Consistency for 8–12 weeks will get you to that finish line strong.
- Don't wait. Don't overthink. Start this week.

Final Words (The Close):

Four principles you'll take with you:

1. **Consistency beats perfection.**
- You don't need a perfect plan or perfect execution. You need consistency.
- One good week repeated 8–12 times = you finish strong.
- Ninety percent adherence is the goal. Perfection is the enemy.

2. **Integration beats isolation.**
- Training fits into your life, not against it.
- When training is integrated (family involved, times are locked, anchors are protected), it's sustainable.
- When training isolates you, it creates resentment and fails.

3. **Adaptation beats rigidity.**
- Life will happen. Work crises, family demands, illness, travel. Plans will bend.
- Flexibility is strength. Adapt and keep moving.
- Plan B is your friend. Use it guilt-free.

4. **Action beats anxiety.**
- Don't overthink. Don't wait for perfect conditions.
- Start this week. Do the training. Anxiety dissolves once you begin.

- By week 2, you'll feel it. By week 8, you'll believe it. By race day, you'll know it.

You can do this.

- You have 8–12 weeks. That's enough time.
- You have a system that works. Thousands of athletes have used it. They finished strong. You will too.
- You have support: family (after the family talk), community (if you join a group), coach (if you reach out).
- You have everything you need.

One last thing:

- The hardest part is starting.
- The second hardest part is week 2–3 (when initial excitement wears off but fitness isn't obvious yet).
- By week 4, it gets easier. By week 8, it's a lifestyle. By week 12, you're ready.
- You've already taken the first step by being here today.
- Now take the second step: go home, do the family talk. Lock in your calendar. Test your zones.
- By next week, you'll be training.
- By race day (8–12 weeks), you'll be ready.

I can't wait to hear your finish line story.